P-block Elements

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Nitrogen Family

Elements (group-15)	Electronic configuration	Nature
$_{7}N^{14}$ -Nitrogen	[He] $2s^2\ 2p^3$	Non-metal
$_{15}P^{31}$ -Phosphorus	[Ne] $3s^2\ 3p^3$	Non-metal
$_{33}As^{75}$ -Arsenic	[Ar] $3d^{10}4s^24p^3$	Metalloid
$_{51}Sb^{122}$ -Antimony	[Kr] $4d^{10}\ 5s^2\ 5s^3$	Metalloid
$_{83}Bi^{209}$ -Bismuth	[Xe] $4f^{14}\ 5d^{10}\ 6s^2\ 6p^3$	Metal

Nitrogen family belongs to group-15. These elements commonly called pnicogens and their compounds are pniciticles.

Atomic and physical properties :

1. Atomic size (A.S.) :

Covalent radius of group-15 elements increase as going down in the group. It is observed that As to Bi small increase in radius due to completely filled d or f-orbitals in heavier members of group.

	N	<	P	<	As	<	Sb	<	Bi
Covalent radius (pm)	70		110		120		140		150

→

2. Ionisation energy (I.E.) :

The I.E. decrease as going down the group due to increase in atomic size. Group-15 elements ionisation energy is much greater than group-14 due to half-filled orbital.

	N	<	P	<	As	<	Sb	<	Bi
(I.E.) [kJ/mole]	1402		1012		947		834		703

→

I.E. ↓

3. Electronegativity (E.N.) :

The E.N. decreases as going down the group with increase in atomic size.

	N	>	P	>	As	>	Sb	=	Bi
E.N.	3.0		2.1		2.0		1.9		1.9

→

E.N. ↓

4. Electron affinity (E.A.) :

The value of electron affinity decreases as going down the group except N and P. Third period elements have high electron affinity due to orbital size is bigger so electron – electron repulsion is minimum and amount of energy release increases.

$$P > N > As > Sb > Bi$$

$$\longrightarrow$$

E.A. ↓

5. Metallic character :

As going down the group, metallic character increases because ionisation energy decreases.

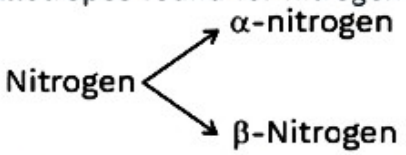
$$\underbrace{N < P}_{\text{non-metals}} < \underbrace{As < Sb}_{\text{Metalloids}} < \underbrace{Bi}_{\text{Metal}}$$

$$\longrightarrow$$

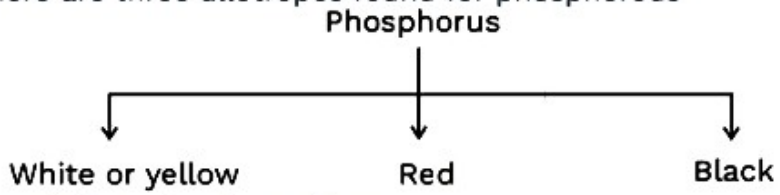
Metallic character ↓

6. Allotropy :

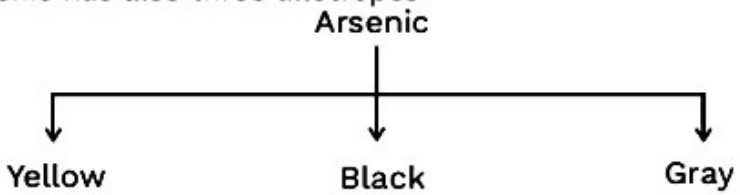
(i) There are two allotropes found for nitrogen



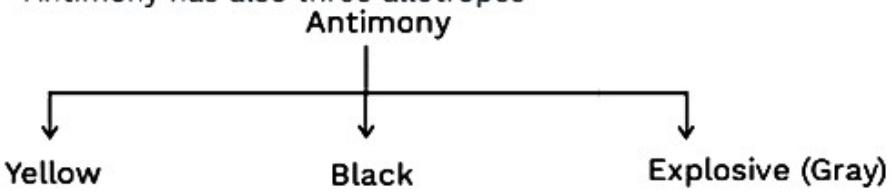
(ii) There are three allotropes found for phosphorous



(iii) Arsenic has also three allotropes



(iv) Antimony has also three allotropes

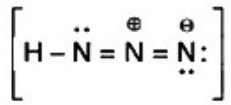


(v) Bismuth does not show allotropy.

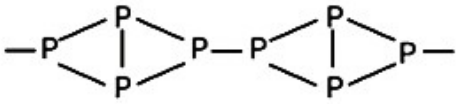
7. Catenation :

Catenation is tendency to form chain. Only N and P show catenation property

→ Nitrogen show in hydrazoic acid (N_3H)



→ Red-P show catenation. (P_4 molecule)



Note :

Most important allotropic form are phosphorous allotropes.

8. Oxidation sate (O.S.) :

The most commonly used oxidation state of group-15 elements are −3, +3 and +5 and as going top to bottom in group, tendency to show positive oxidation state increases due to "inert pair effect".

+5 oxidation state decreases down to group and +3 oxidation sate increases down the group.

$\boxed{Bi^{+3} > Sb^{+3} > As^{+3}}$

$\boxed{Bi^{+5} > Sb^{+5} > As^{+5}}$

→ Nitrogen shows −3 to +5 all oxidation state in different compounds.

Molecules	O.S.
NH_3	−3
N_2H_4	−2
NH_3−OH	−1
N_2	0
N_2O	+1
NO	+2
HNO_2	+3
NO_2	+4
HNO_3	+5

→ Maximum oxidation state :
= (Number of electron in valence shell)

→ Minimum oxidation state :
= (Number of electron in valence shell -8)

Q.1 **PF_5 is known but NF_5 is not.**

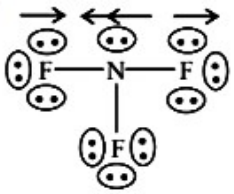
Sol. PF_5 is known but NF_5 is unknown because in nitrogen 'd' sub shell is absent. Thus it can't form extra bond with lone pair of Fluorine. Thus NF_5 don't possible.

Q.2 **NF_3 is stable but NCl_3 and NI_3 are explosives.**

Sol. In NF_3, there is strong pp-pp bonding due to large electronegativity difference between N(3) and F(4). On the other hand there is weaker pp-pp bonding in NCl_3 and NI_3 hence, these compounds are unstable and explosive.

Q.3 **The experimentally determined N – F bond length in NF_3 is greater than the sum of single bond covalent radii of N and F.**

Sol.



Repulsion of lone pair of N and Fluorine

In NF_3 molecules, bond length between N and F is large due to repulsion between non bonded pair of electrons due to its smaller size and high electron density.

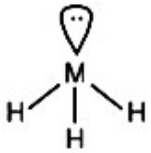
Chemical properties

1. Hydrides :

(i) All members form hydrides (MH_3) of group-15 elements.

NH_3	PH_3	AsH_3	SbH_3	BiH_3
Ammonia	Phosphine	Arsine	Stibine	Bismuthine

→ Every hydride have sp^3 hybridisation on centrad atom.
→ Shapes of hydrides are pyramidal due to presence of lone-pair.



(M = metal)

(ii) Order of thermal stability :
As going down the group thermal stability decreases, due to their bond dissociation energy.

$$NH_3 > PH_3 > AsH_3 > SbH_3 > BiH_3$$

Thermal stability ↓

(iii) Reducing nature :
As going down the group reducing nature increases. Ammonia is only mild reducing agent while BiH_3 is strong reducing agent among all hydrides.

$$NH_3 < PH_3 < AsH_3 < SbH_3 < BiH_3$$

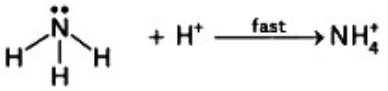
Reducing nature ↑

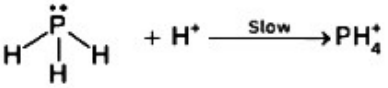
(iv) Basic nature :
All hydrides act as Lewis base due to presence of lone pair.
→ Density of lone pair decreases down to the group due to increase in size so basic properties decreases from top to bottom.

$$\ddot{N}H_3 > \ddot{P}H_3 > \ddot{A}sH_3 > \ddot{S}bH_3 > \ddot{B}iH_3$$

Basic character ↓

Note : Tendency to form ammonium ion $\left(NH_4^+\right)$ is higher than phosphonium ion $\left(PH_4^+\right)$





(v) Bond Angle :

As electronegativity of central atom decreases down the group, bond angle also decreases.

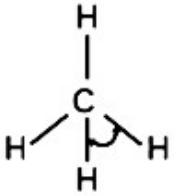
Bond angle :

NH_3	PH_3	AsH_3	SbH_3	BiH_3
106.5°	93.5°	91.5°	91.3°	90°

Note : Concept of bond angle :

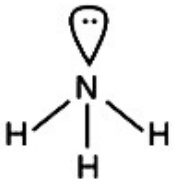
1. Concept – I

$$\text{Bond angle} \propto \frac{1}{\text{number of lone pair on central atom}}$$



109° 28'

sp^3



106.5°

sp^3



104.5°

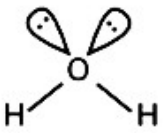
sp^3

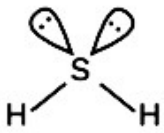
Order of repulsion :
l.p. – l.p. > l.p. – B.P. > B.P. – B.P.
l.p. = lone pair
B.P. = Bond pair

2. Concept-II

If hybridisation and lone pair are same then

Bond angle ∝ Electronegativity of central atom





104.5° 92.5°

sp^3 sp^3

Order of bond angle :
$H_2O > H_2S > H_2Se > H_2Te$

3. **Concept – (III)**
If hybridisation on central atom is different then bond angle is different.

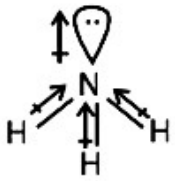
sp >	sp^2 >	sp^3
180°	120°	109°8'

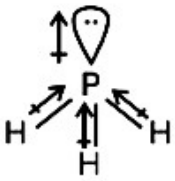
(vi) **Dipole moment :**
As electronegativity of central atom decreases from top to bottom dipole moment decreases:

$NH_3 > PH_3 > AsH_3 > SbH_3 > BiH_3$

→

Dipole moment ↓





(vii) **Boiling point :**
Boiling point of NH_3 is higher than PH_3 because NH_3 has intra molecular H–bonding.
→ From PH_3 to BiH_3 as molecular mass increases Vander waal force of attraction increases. So boiling point increases :

PH_3 <	AsH_3 <	NH_3 <	SbH_3 <	BiH_3
185.5 K	210 K	238.5 K	254 K	290 K

(vii) Covalent character :

$$\Delta E.N. \times \text{ionic character} \propto \frac{1}{\text{Covalent character}}$$

$\Delta E.N. \Rightarrow$ Difference in electronegativity.

As going top to bottom, electronegativity decrease, sharing tendency of electron increases so covalent character increases.

$$NH_3 < PH_3 < AsH_3 < SbH_3 < BiH_3$$

$\longrightarrow$

Covalent character $\uparrow$

Q.4 **Among this incorrect increasing order of bond order :**

(A) $NH_3 < H_2O < H_2S < CH_4$ **(B)** $CH_4 < NH_3 < H_2O < H_2S$

(C) $H_2S < H_2O < NH_3 < CH_4$ **(D)** $H_2O < H_2S < NH_3 < CH_4$

Sol. (C)

Q.5 **The correct order of thermal stability of hydrides of group 15 is -**

(A) $NH_3 > PH_3 > AsH_3 > BiH_3 > SbH_3$

(B) $NH_3 > PH_3 > AsH_3 > SbH_3 > BiH_3$

(C) $NH_3 < PH_3 < SbH_3 > AsH_3 > BiH_3$

(D) $BiH_3 > SbH_3 > AsH_3 > PH_3 > NH_3$

Sol. **[B]** Thermal stability of hydrides of group 15 is

NH_3	>	PH_3	>	AsH_3	>	SbH_3	>	BiH_3
1300°C		440°C		280°C		150°C		room temperature.

Q.6 **Arrange the group 15 hydrides in the increasing order of H–M–H bond angle.**

(A) $SbH_3 < AsH_3 < PH_3 < NH_3$ **(B)** $NH_3 < PH_3 < AsH_3 < SbH_3$

(C) $NH_3 < SbH_3 < AsH_3 < PH_3$ **(D)** $NH_3 < PH_3 < SbH_3 < AsH_3$

Sol. **[A]** Bond angle of group 15 hydrides is in order of

SbH_3	<	AsH_3	<	PH_3	<	NH_3
91.3°		91.8°		93.8°		107.2°

Q.7 **NH_3 (excess) + Cl_2 ⟶ ..... + ...**

Sol. NH_4Cl and N_2

Q.8 **Arrange these in correct order of increasing order of bond angle.**

(A) $CO_2 < BF_3 < CH_3$ **(B)** $BF_3 < CO_2 < CH_4$

(C) $CH_4 < BF_3 < CO_2$ **(D)** None

Sol. (C)

Q.9 **Arrange following in correct order of increasing order of bond angle :**

(A) $BiCl_3 < SbCl_3 < PCl_3 < NCl_3 < BCl_3$

(B) $BCl_3 < NiI_3 < PCl_3 < SbCl_3 < BiCl_3$

(C) $NCl_3 < PCl_3 < SbCl_3 < BiCl_3 < BCl_3$

(D) None

Sol. (A)

2. Halides of N-family :

(i) Tri halides :

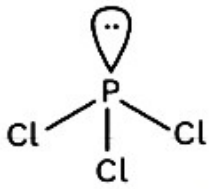
(a) All tri halides have sp^3 hybridisation on central atom due to presence of lone pair

the structure is pyramidal.

Example : PCl_3

P = $\boxed{\uparrow\downarrow}$ $\boxed{1}\boxed{1}\boxed{1}$

$3s^2$ $3P^3$



Hybridisation → sp^3

Structure → Pyramidal

Geometry → Tetrahedral

(b) All trihalide except NCl_3, NBr_3, NI_3 are stable.
These are unstable because :
- Low polarity of N–X bond
- Large size difference

(c) In trihalides of nitrogen, as the electronegativity of halogen atom decreases, availability of lone-pair on nitrogen increases, so Lewis base strength increases.

$$\ddot{N}F_3 < \ddot{N}Cl_3 < \ddot{N}Br_3 < \ddot{N}I_3$$

Basic strength ↑

Note : Trihalide of phosphorus
PX_3 {X = F, Cl, Br, I}

Preparation :

(i) $P_4 + 6Cl_2 \xrightarrow{\Delta} 4PCl_3$
(white P, dry chlorine)

(ii) $P_4 + 8SOCl_2 \xrightarrow{\Delta} 4PCl_3 + 4SO_2 + 2S_2Cl_2$
(white P, thionly chloride)

Properties :

(i) It is colourless and oily liquid and in the presence of moisture it can be hydrolyses.

$PCl_3 + 3H_2O \longrightarrow H_3PO_3 + 3HCl$

(ii) Reaction with organic compounds (having –OH group)

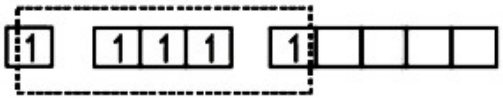
$PCl_3 + 3CH_3COOH \longrightarrow H_3PO_3 + 3CH_3COCl$

(iii) **Penta halides :**

→ Penta halides for nitrogen can not formed due to non availability of d-orbitals.

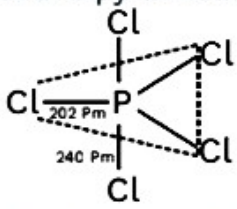
→ Penta halides of phosphorus (P) → PCl_5
All penta halides have sp^3d hybridisation

$PCl_5 \Rightarrow P =$ 

p* Excited state 

Hybridisation → sp^3d

Structure → trigonal bipyramidal

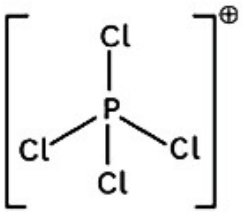


→ Three equatorial bond are equal and strong

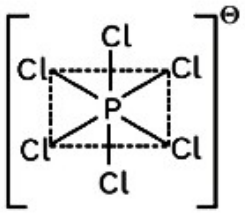
→ Two axial bonds are longer and reactive

Note : X-rays studies shows that in solid sate PCl_5 exist in ionic form.

Solid PCl_5 exists as ⇒ $[PCl_4]^+[PCl_6]^-$



sp^3
Tetrahedral



sp^3d
Octahedral

Preparation of PCl_5 :

(i) $\underset{\text{White P}}{P_4} + 10\underset{\text{dry } Cl_2}{\underset{\text{Excess of}}{Cl_2}} \longrightarrow 4PCl_5$

(ii) $P_4 + 10SO_2Cl_2 \longrightarrow 4PCl_5 + 10SO_2$

Properties of PCl_5 :

(i) PCl_5 is yellowish white powder. In presence of moist air it hydrolysis in $POCl_3$ and finally get converted in phosphonic acid-

$PCl_5 + H_2O \longrightarrow POCl_3 + 2HCl$

$POCl_3 + 3H_2O \longrightarrow H_3PO_4 + 3HCl + \frac{1}{2}O_2$

(ii) It decomposes on strong heating.

$$PCl_5 \xrightarrow[\text{heating}]{\text{strong}} PCl_3 + Cl_2$$

(iii) Reaction with organic compounds (having –OH group).

$$C_2H_5OH + PCl_5 \longrightarrow C_2H_5Cl + PCl_3 + HCl$$

$$CH_3COOH + PCl_5 \longrightarrow C_2H_5COCl + POCl_3 + HCl$$

(iv) Reaction with metals.

$$2Ag + PCl_5 \longrightarrow 2AgCl + PCl_3$$

$$Sn + 2PCl_5 \longrightarrow SnCl_4 + 2PCl_3$$

Q.10 **Which of following are not known–**

(A) PH_5 **(B) PI_5**

(C) NCl_5 **(D) NF_5**

Sol. **[A,B,C,D]**

In Nitrogen (d) subshell is absent thus it can't form NCl_5, NF_5 . PH_5 and PI_5 doesn't possible.

Q.11 **Which of the following conclusions are correct?**

(A) The enthalpy of formation of NF_3 is exothermic whereas enthalpy of for mation of NCl_3 is endothermic

(B) The enthalpy of formation of NF_3 and NCl_3 both are exothermic

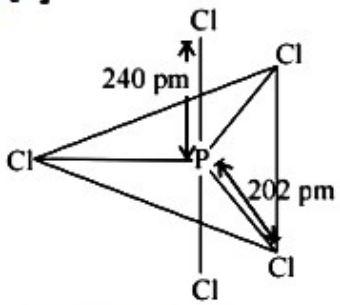
(C) N–N bond distance in N_2O_4 is greater than the N–N usual single bond distance

(D) N–N bond distance in N_2O_4 is smaller than the N–N usual single bond distance

Sol. **[A,C]**

Q.12 **Assertion : PCl_5 molecules has two bond angles of Cl–P–Cl as 90° and 120° as well as two P–Cl bonds are weaker then rest three P–Cl bonds.**
Reason : P in PCl_5 shows sp^3 d hybridisation leading to trigonal bi-pyramid geometry. Also two P–Cl bonds at axial positions give less stable bonds.

Sol. **[A]**



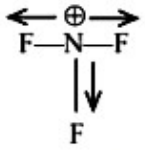
Cl—P—Cl = 90°
Cl—P—Cl = 120°
sp^3d (Trigonal bipyramidal)
Axial bond length is 240 pm and equatorial bond length, is 202 pm thus axial bond (Cl—P) is less stable.

Q.13 **Assertion : The basic nature of trihalides of nitrogen decreases from NF_3 to NI_3.**
Reason : The halogens develop + ve charge on N atom and thus more +ve charge is developed in NF_3 and therefore tendency to lose electron pair decreases.

Sol. **[D]**

NF_3 NCl_3 NBr_3 NI_3

Basic nature increases



Due to high electronegativity of fluorine it creates more positive charge on Nitrogen thus it's electron donating tendency decreases

Q.14 **$[PBr_6]^-$ anion is not formed but $[PCl_6]^-$ anion exist.**

Sol. PBr_5 in solid there is ionic and has a structure $[PBr_4]^+ Br^-$ doesn't form anion $[PBr_6]^-$ but PCl_5 in solid state exist in form of $[PCl_5]^+$ and $[PCl_6]^-$ anion.

Q.15 **Solid PCl_5 exists as–**

(A) PCl_5 **(B) PCl_4^+ & PCl_6^-**

(C) PCl_6^- **(D) PCl_4^+**

Sol. **[B]**

X-Ray studies have revealed that solid PCl_5 consists of ion lattics i.e. tetrahedral $[PCl_4]^+$ cation and octahedral $[PCl_6]^-$ anions.

3. Oxides of N-family :

All the elements of N-family form X_2O_3, X_2O_4 and X_2O_5 type of oxides (where X = N,P, As,Sb,Bi).

→ Oxide are more acidic which have higher oxidation state of the element than that of lower oxidation state.

→ If centre atom is same than-

Oxidation state on central atom $\propto$ Acidic strength of oxide

Example :

$$\overset{+3}{N_2O_3} < \overset{+4}{N_2O_4} < \overset{+5}{N_2O_5}$$

→ If oxidation state is same than as electronegativity of central atom decreases acidic strength decreases.

Example :

$$\overset{+3}{N_2O_3} > \overset{+3}{P_2O_3} > \overset{+3}{As_2O_3} > \overset{+3}{Sb_2O_3} > \overset{+3}{Bi_2O_3}$$

Q.16 **Which oxide has lowest acid strength ?**

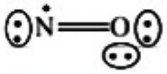
(A) N_2O_5 **(B) P_2O_5** **(C) As_2O_3** **(D) Sb_2O_3**

Sol. **(D)**

Q.17 **NO is paramagnetic in gaseous state but diamagnetic in liquid or solid state.**

Sol. **True**

The molecule contain an odd electron which makes the gaseous nitric oxide paramagnetic



In the liquid and solid states NO is known to form a loose dimer N_2O_2 thus it is diamagnetic.

Q.18 **Which reaction can be used to prepare phosphorus acid–**

(A) $P_2O_3 + H_2O \xrightarrow{20^\circ C}$ **(B)** $P_2O_3 + H_2O \xrightarrow{25^\circ C}$

(C) $P_2O_3 + H_2O \xrightarrow{80^\circ C}$ **(D)** $P + \text{conc. } HNO_3 \longrightarrow$

Sol. **[A]**

Phosphorus acid is obtained dissolving phosphorus trioxide in water (at low temperature)

$2P_2O_3 + 6H_2O \xrightarrow{20^\circ C} 4H_3PO_3$

Q.19 **Arrange the oxides of group 15 elements in decreasing order of their acidity–**

(A) $N_2O_5 > P_2O_5 > As_2O_5 > Sb_2O_5 > Bi_2O_5$

(B) $Bi_2O_5 > Sb_2O_5 > As_2O_5 > P_2O_5 > N_2O_5$

(C) $P_2O_5 > N_2O_5 > As_2O_5 > Sb_2O_5 > Bi_2O_5$

(D) $N_2O_5 > Bi_2O_5 > P_2O_5 > As_2O_5 > Sb_2O_5$

Sol. **[A]**

Central atoms electronegativity $\propto$ Acidic nature

$N_2O_5 > P_2O_5 > As_2O_5 > Sb_2O_5 > Bi_2O_5$

Q.20 **Which of the following is least acidic–**

(A) P_4O_6 **(B) P_4O_{10}** **(C) As_4O_6** **(D) As_4O_{10}**

Sol. **[C]**

Acidic nature $\propto$ oxidation state of central atom $\propto$ electronegativity of centre atom $\propto$ Electronegativity of attached side atom

As_4O_6	As_4O_{10}
→ Centre atom E.N. = Same → attached side oxygen atom = 6	Same 10

Thus As_4O_6 is less acidic as compare to As_4O_{10}

Compounds of Nitrogen

1. Dinitrogen (Nitrogen) N_2 :

Dinitrogen comprises 78% by volume of earth atmosphere.

Preparation:

(i) Laboratory Method :

By treating ammonium chloride with sodium nitrite, N_2 collected by the down-ward displacement.

$$NH_4Cl + NaNO_2 \xrightarrow{\Delta} NH_4NO_2 + NaCl$$

$$NH_4NO_2 \xrightarrow{\Delta} N_2 + 2H_2O$$

(ii) Thermal decomposition of ammonium dichromate :

$$(NH_4)_2Cr_2O_7 \xrightarrow{\Delta} N_2 + 4H_2O + Cr_2O_3$$

(iii) To obtain very pure nitrogen, thermal decomposition of sodium or barium azide takes place.

$$Ba(N_3)_2 \xrightarrow{\Delta} Ba + 3N_2$$

Properties of Dinitrogen :

(I) It is colourless and odouless gas, which is less soluble in water it is non-toxic in nature.

(II) It is chemically inert at room temperature due to high bond enthalpy of N ≡ N.

(III) At high temperature it is reactive. It react with metals and form nitrides.

Examples :

(i) $3Mg + N_2 \xrightarrow[\text{temperature}]{\text{High}} Mg_3N_2$

(ii) $3Ca + N_2 \xrightarrow{\Delta} Ca_3N_2$

(IV) At very high temperature it combine with $O_2(g)$ to form nitric oxide gas.

$$N_2(g) + O_2(g) \underset{\Delta, Fe}{\overset{2000K}{\rightleftharpoons}} 2NO(g)$$

(V) Haber's process (preparation of ammonia)

$$N_2(g) + 3H_2(g) \overset{773K}{\rightleftharpoons} 2NH_3(g)$$

(vi) Reaction with CaC_2 and BaC_2 at 1100°C.

$$CaC_2 + N_2 \xrightarrow{\Delta} \underbrace{CaCN_2 + C}_{\text{(Nitrolime)}}$$

$$BaC_2 + N_2 \xrightarrow{\Delta} BaCN_2$$

Uses :

(1) In metallurgy of take the environment inert.

(2) In production of electric bulbs.

Q.21 **Nitrogen combines with metals to form -**

(A) Nitrites **(B) Nitrates**

(C) Nitrosyl chlorides **(D) Nitrides**

Sol. **[D]** Highly electropositive metals like lithium or magnesium burning in the atmosphere of dinitrogen to form metalic nitrides

$6Li(s) + N_2(g) \rightarrow 2Li_3N$

lithium nitride

Oxides of nitrogen

Nitrogen forms different oxides. These oxides exhibit p_π p_π multiple bonding between nitrogen and oxygen.

Properties of oxides of nitrogen

Compounds	Formula	Preparation methods	Physical and chemical properties
Dinitrogen oxide [Nitrogen(I) oxide]	N_2O	$NH_4NO_3 \xrightarrow{\text{Heat}} N_2O + 2H_2O$	colourless gas, neutral
Nitrogen monoxide [Nitrogen(II) oxide]	NO	$NO_3^- + 3Fe^{2+} + 4H^+ \rightarrow NO + 3Fe^{3+} + 2H_2O$ $2HNO_2 + 2I^- + 2H^- \rightarrow 2NO + I_2 + 2H_2O$ $KNO_2(l) + KNO_3(l) + Cr_2O_3(s) \rightarrow 2K_2CrO_4(s,l) + 4NO$	colourless gas, neutral
Dinitrogen trioxide [Nitrogen(III) oxide	N_2O_3	$2NO + N_2O_4 \xrightarrow{250K} 2N_2O_3$	blue solid/liquid, acidic
Nitrogen dioxide [Nitrogen(IV) oxide]	NO_2	$2Pb(NO_3)_2 \xrightarrow{673K} 4NO_2 + 2PbO + O_2$ $Cu + 4\ HNO_3\ (conc.) \rightarrow Cu(NO_3)_2 + 2NO_4 + 2H_2O$	brown gas, acidic
Dinitrogen tetroxide [Nitrogen(IV) oxide]	N_2O_4	$2NO_2 \underset{\text{Heat}}{\overset{\text{Cool}}{\rightleftharpoons}} N_2O_4$	colourelss solid/liquid, acidic
Dinitrogen pentoxide [Nitrogen(IV) oxide]	N_2O_5	$4\ HNO_3 + P_4O_{10} \rightarrow 4HPO_3 + 2N_2O_5$ $2NO_2 + O_3 \rightarrow N_2O_5 + O_2$	colourless solid, acidic

2. Nitrous oxide (N_2O) [Laughing gas] :

Preparation :

By heating of ammonium nitrate

$$NH_4NO_3 \xrightarrow{\Delta} N_2O + H_2O$$

Properties :

(i) It is colourless gas which is neutral towards litmus paper.

(ii) It is used in the prepration of diffrent azides

$$N_2O + 2NaNH_2 \longrightarrow NaN_3 + NH_3 + NaOH$$

When it smell by humen it causes hysterical laughing.

Structure of N_2O

$$[:\overset{\ominus}{N} = \overset{\oplus}{N} = \ddot{O}:] \longleftrightarrow [\ddot{N} \equiv \overset{\oplus}{N} - \overset{\ominus}{\ddot{O}}:]$$

Note : N_2O is isoelectronic and isostructural with CO_2. Both have 22 electrons and linear structure.

$N_2O = 14 + 8 = 22$, $CO_2 = 6 + 16 = 22$

Uses : Mixture of N_2O and O_2 is used as antishetic.

2. Nitric oxide (NO) :

Preparation :

(i) From electric Arc-process

$$N_2 + O_2 \xrightarrow[\text{Arc, 200K}]{\text{Electric}} 2NO$$

(ii) From Ostwald process :

$$NH_3 + O_2 \xrightarrow[\substack{\text{6–7 atm}\\ \text{Pt-foil}}]{\text{750–900°C}} NO + H_2O$$

Properties :

(i) It is colourless gas and neutral towards litmus.

(ii) When it is heated in presence of O_2, it gives brown vapour of NO_2.

$$NO + O_2 \xrightarrow[\text{temperature}]{\text{High}} \underset{\text{brown vapour}}{NO_{2(g)}}$$

(iii) Brown ring test : when it is heated with aqueous solution ferrous sulphate give brown colour complex :

$$[Fe(H_2O)_6]SO_4 + NO \longrightarrow \underset{\substack{\text{Hydrated nitrosyl complex}\\ \text{(Brown colour)}}}{[Fe(H_2O)_5NO]SO_4} + H_2O$$

(iv) It is paramagnetic in nature, due to odd electron bond.

$$:N=\ddot{O}: \longleftrightarrow :\dot{N}\equiv\ddot{O}:$$

Uses : It is used in lead chamber process as catalyst

$$SO_2 + H_2O + [O] \xrightarrow{NO} H_2SO_4$$

3. Dinitrogen trioxide (N_2O_3) :

Preparation :

$$2NO + N_2O_4 \xrightarrow{250K} 2N_2O_3$$

Properties :

(i) It is blue acidic liquid.

(ii) When it is condensed at 0–30°C temperature then it is found as blue liquid but at room temperature it gives vapour of NO and NO_2.

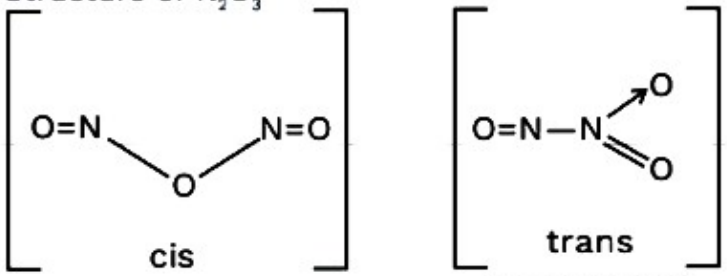
$$N_2O_3 \xrightarrow{\text{room temperature}} \underbrace{NO + NO_2}_{\text{Brown vapours}}$$

(iii) By reaction with water it gives nitrous acid, So also called anhydride of nitrous acid.

$$N_2O_3 + H_2O \longrightarrow 2HNO_2$$

$$\begin{matrix} H-NO_2 \\ HO-NO \end{matrix} \xrightarrow{-H_2O} N_2O_3$$

4. Structure of N_2O_3



cis

(More stable)

trans

(Less stable)

4. Nitrogen dioxide (NO_2) or N_2O_4 :

Proration :

By heating lead nitrate crystals

$$2Pb(NO_3)_2 \xrightarrow{673K} 4NO_2 + 2PbO + O_2$$

Properties :

(i) NO_2 is brown colour gas, acidic in nature.

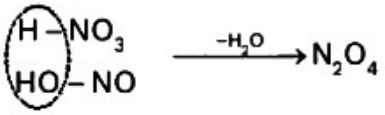
(ii) When it is cooled to –11°C then it stern in to colourless liquid.

$$\underset{\text{Brwon colour gas}}{NO_2} \underset{}{\overset{-11^\circ C}{\rightleftharpoons}} \underset{\text{Colour less liquid}}{N_2O_4}$$

(iii) By reaction with water, it gives nitrous acid and nitric acid.

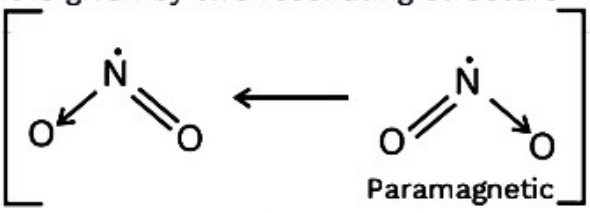
$$2NO_2 + H_2O \longrightarrow HNO_2 + HNO_3$$

- It is mixed on anhydride of HNO_2 and HNO_3.

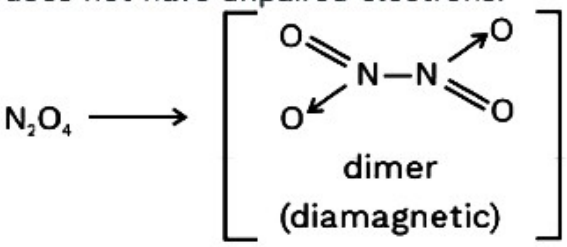


(iv) **Structure of N_2O_4**

It is given by two resonating structure



At low temperature it converts in N_2O_4 which is diamagnetic in nature because it does not have unpaired electrons.



Dinitrogen penta oxide (N_2O_5) :

Preparation :

(i) $4HNO_3 + P_4O_{10} \longrightarrow 4HPO_3 + 2N_2O_5$

(ii) $AgNO_3 + Cl_2 \longrightarrow AgCl + N_2O_5 + O_2$

Properties :

(i) It is colourless acidic solid

(ii) By treating with water, it gives nitric acid

$N_2O_5 + H_2O \longrightarrow 2HNO_3$

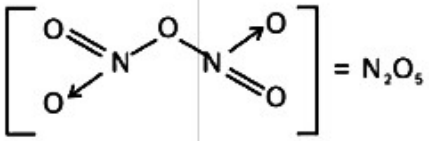
(iii) When reacted with solution of NaCl, then ionic reaction occur which proves that solid N_2O_5 is ionic in nature

$N_2O_5 + NaCl \longrightarrow NaNO_3 + NO_2Cl$

$\boxed{\text{Solid } N_2O_5 = (NO_2^+)(NO_3^-)}$

Nitronium nitrate

(iv) Structure of N_2O_5



$= N_2O_5$

Q.22 On heating $Pb(NO_3)_2$ produces a pale yellow gas which on strong heating produces–

(A) N_2O_4, brown gas **(B) N_2, colourless gas**

(C) NO_2, brown gas **(D) N_2O_3 , pale blue gas**

Sol. [C]

$2Pb(NO_3)_2 \rightarrow 2PbO + 4NO_2 + O_2$

Brown gas

Q.23 Nitrogen dioxide can not be obtained by heating -

(A)KNO_3 **(B) $Pb(NO_3)_2$** **(C) $Cu(NO_3)_2$** **(D) $AgNO_3$**

Sol. [A] NO can be obtained by heating heavy metals nitrates. KNO_3 is not the nitrate of heavy metal.

Q. 24 **Light blue colour of nitrous acid is due to dissolved–**

(A) O_2 **(B) N_2** **(C) N_2O** **(D) N_2O_3**

Sol. **[D]**

$N_2O_3 + H_2O \rightarrow 2HNO_2$

A solution of nitrous acid may also be prepared by dissolving N_2O_3 in water aqueous solution of nitrous acid is light blue colour.

Q.25 **Which of the following is/are paramagnetic–NO_2, NO, N_2O_4, N_2O_2, N_2O_5**

(A) Only NO_2 **(B) NO_2, NO**

(C) NO, N O_2, N_2O_5 **(D) All are paramagnetic**

Sol. **[B]**

It has unpaired electron thus it is paramagnetic [compound has odd number of e^-]

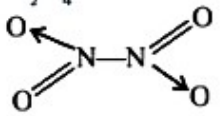
$NO_2 = 23e^-$ ← Odd electron Paramagnatic

$NO = 15\ e^-$ ←

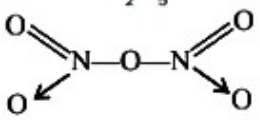
$N_2O_4 = 46e^-$ ← even number of electron diamagnatic

$N_2O_5 = 54\ e^-$ ←

N_2O_4



N_2O_5



In N_2O_4 and N_2O_5 all electron of nitrogen is bonded thus it is diamagnatic.

Oxy acids of nitrogen

1. Nitrous acid (HNO_2) :

Preparation :

(i) From Barium nitrate

$$Ba(NO_2)_2 + H_2SO_4 \longrightarrow 2HNO_2 + BaSO_4$$

(ii) By reaction of an equimolar mixture of NO and NO_2 with water

$$NO + NO_2 + H_2O \longrightarrow 2HNO_2$$

Properties :

(i) It is weak acid, exist in liquid state and unstable.

(ii) By increasing concentration, it will decompose into nitric acid.

$$3HNO_2 \longrightarrow HNO_3 + 2NO + H_2O$$

(iii) Nitrous acid and nitrites are good oxidizing agent.

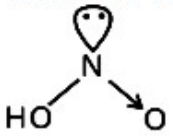
Example:

$2KI + 2HNO_2 + 2HCl \longrightarrow 2H_2O + 2NO + 2KCl + I_2$

(iv) They also act as reducing agents with respect to strong oxidizing agents like $KMnO_4$.

$2KMnO_4 + 5KNO_2 + 6HCl \longrightarrow 2MnCl_2 + 5KNO_3 + 3H_2O + 2KCl$

(v) Structure of HNO_2



Hybridisation = sp^2

Bond Angle < 120°

2. Nitric acid (HNO_3)

Nitric acid is also called as **Aqua Fortis**. it act as dehydrating agent.

Preparation :

I. Ostwald Process :

Step-1 Catalytic oxidation of NH_3 by atmospheric oxygen

$$4NH_3(g) + 5O_2(g) \text{ (from air)} \xrightarrow[500K,\ 9bar]{\text{Pt/Rh gauge catalyst}} 4\ NO(g) + 6H_2O(g)$$

Step-2 Nitric oxide formed combine with oxygen to give NO_2

$$2NO(g) + O_2(g) \rightleftharpoons 2NO_2(g)$$

Step-3 NO_2 dissolve in water to give Nitric acid

$$3NO_2(g) + H_2O(l) \longrightarrow 2HNO_3(aq) + NO(g)$$

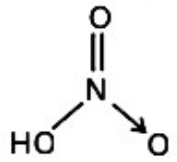
Obtained aqueous $AgNO_3$ can be concentrated upto 68% by mass by distillation.

Properties :

Physical properties :

(i) It is colourless liquid.

(ii) In gaseous state, HNO_3 exist as a planar molecule.



Hybridisation = sp^2

In aqueous medium nitric acid is strong acid

(iii) Concentrated nitric acid is a strong oxidizing agent and react with most of the metals except noble metals (gold & Pt).

$3Cu + 8HNO_3 \xrightarrow{\text{(dil.)}} 3Cu(NO_3)_2 + 2NO + 4H_2O$

$Cu + 4HNO_3 \xrightarrow{\text{(conc.)}} Cu(NO_3)_2 + 2NO_2 + 2H_2O$

$4Zn + 10HNO_3 \xrightarrow{\text{(dil.)}} 4Zn(NO_3)_2 + N_2O + 5H_2O$

$Zn + 4HNO_3 \rightarrow Zn(NO_3)_2 + 2NO_2 + 2H_2O$

Some of metals like Cr, Al do not dissolve in concentrated nitric acid due to protection film of oxide on the surface.

(iv) Concentrated nitric acid also oxidises non-metals and their compounds.

$I_2 + 10HNO_3 \longrightarrow 2HIO_3 + 10NO_2 + 4H_2O$

$C + 4\ HNO_3 \longrightarrow CO_2 + 2H_2O + 4NO_2$

$S_8 + 48\ HNO_3(\text{concentrated}) \longrightarrow 8H_2SO_4 + 48\ NO_2 + 16\ H_2O$

$P_4 + 20\ HNO_3\ (\text{concentrated}) \longrightarrow 4H_3PO_4 + 20NO_2 + 4H_2O$

Uses:

(i) It is used in manufacturing of fertilizers.
(ii) It is used in manufacturing of explosives.
(iii) It is used in metallurgical process.
(iv) It is used in rocket fuels as oxidizer.

3. Hyponitrous Acid : ($H_2N_2O_2$)

Preparation :

$N_2O + H_2O \longrightarrow H_2N_2O_2$

Structure :

$[HO - \ddot{N} = \ddot{N} - OH]$

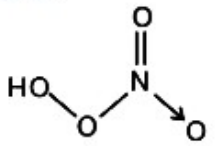
Hybridisation on both nitrogen in sp^2

4. Per Nitric acid (HNO_4) :

Preparation :

$H_2O_2 + N_2O_5 \longrightarrow HNO_4 + HNO_3$

Structure :



Hybridisation = sp^2

5. Per Nitrous Acid (HOONO) :

Preparation :

$H_2O_2 + ON(OH) \longrightarrow ON(OOH) + H_2O$
(HNO_3)

Structure :

$[H - O - O - \ddot{N} = O]$

Hybridisation is sp^2

Q.26 **Concentrated HNO_3 turns yellow in sun light.**

Sol. HNO_3 decomposes to form yellow N_2O_4 in sun light

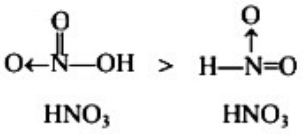
$$4HNO_3 \xrightarrow{\text{Sunlight}} 2H_2O + 2N_2O_4 + O_2$$

Q.27 **Nitrate and nitrite cannot be distinguished by the action of conc. H_2SO_4.**

Sol. Because both liberate brown fumes of NO_2 on heating with conc. H_2SO_4

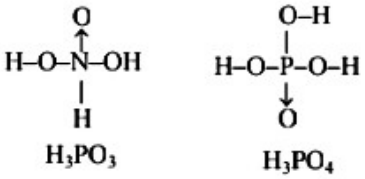
Q.28 **Although HNO_3 is stronger than HNO_2, the H_3PO_3, H_3PO_4 and are of approximately same strength.**

Sol.



HNO_3 $\quad$ HNO_3

HNO_3 has one acidic hydrogen but HNO_2 has no. Thus HNO_3 is more acidic than HNO_2.



H_3PO_3 $\quad$ H_3PO_4

it is dibasic $\quad$ it is tribasic

$K_1 = 10^{-1}$, $\quad K_2 = 10^{-7}$ $\quad$ $K_1 = 10^{-1}$ $\quad K_2 = 10^{-7}$

$K_3 = 10^{-10}$

In case of H_3PO_4 K_2, K_3 is very less thus both have same strength.

Q.29 **Copper dissolves in HNO_3 but not in HCl.**

Sol. HNO_3 is an oxidising agent which converts copper into CuO. Which dissolves in HNO_3 in the form of copper nitrate

Phosphorus and its oxy Acids

(A) Phosphorus has three allotropes.

(1) White or yellow phosphors (P_4) :

Preparation : By electric furnace method

$2Ca_3(PO_4)_2 + 10C + 6SiO_2 \longrightarrow 6CaSiO_3 + 10CO + P_4(s)$
(from bone ash)

Properties :

(i) It is toxic in nature.

(ii) Its burning temperature is low, so it catches fire easily that's why it is kept in water.

(iii) It is insoluble in water but soluble in CS_2.

(iv) When it is placed in dark it gives light, it is called "Phosphorescence". In dark it burn slowly and emits its energy in form of light.

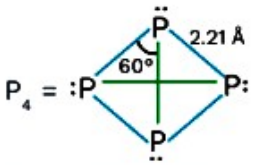
(v) It gives PH_3(phosphene) gas by reaction with NaOH.

$$P_4 + NaOH + H_2O \longrightarrow \underset{\text{Sodium hypophosphite}}{NaH_2PO_2} + PH_3$$

(vi) When P_4 is heated in presence of O_2, it gives oxides :

$$P_4 + 5O_2 \longrightarrow P_4O_{10}$$

(vii) Structure :



- Every phosphorus is sp^3 hybridised.
- Every phosphorus have pyramidal structure but whole P_4 molecule has tetrahedral structure.

Q.30 Which one of them is incorrect.

(A) It has 4 P–P bond **(B) It has 6 P–P bond**

(C) P–P–P bond angle is 60° **(D) Every P-atom have lone-pair**

Sol. (A)

2. Red phosphorus :

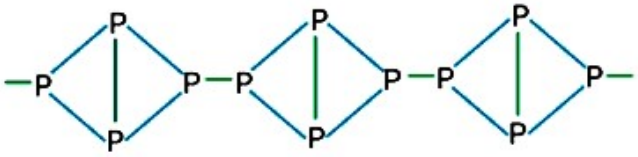
Preparation :
Red phosphorous is made by heating white phosphours in the atmosphere of CO_2 coal gas. This red phosphorus may still contain same white phosphorus which is removed by boiling with NaOH.

$$P_4 + 3NaOH + 3H_2O \longrightarrow PH_3(g) + 3NaH_2PO_2$$

Properties :

(i) It is toxic in nature.
(ii) It burning temperature is high
(iii) It is insoluble in water and CS_2 both.
(iv) It does not react with base.
(v) It does not show phosphorescence.
(vi) When heated in presence of O_2, it also oxidise and give P_2O_5.
(vii) Structure :

P_4 molecule combine together and form a polymer type structure of red phosphorus.



Red Phosphorus

3. Black phosphorus :

Preparation : Black phosphorus have two forms.

(i) **α-Black phosphorus :** When red P heated in a sealed tube at 803 K.

$$\text{Red P} \xrightarrow[\text{Sealed tube}]{803K} \alpha\text{-black P}$$

(ii) **β-black phosphorus :** By heating white phosphorus at 473 K under high pressure. It does not oxidise up to 673K.

$$\text{White P} \xrightarrow[\text{high pressure}]{473K} \beta\text{-black P}$$

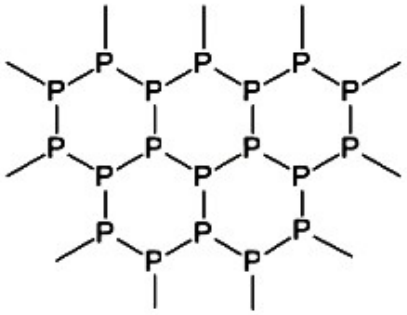
Properties :

(i) It is used in semiconductors

(ii) It does not oxidise in air

(iii) β-black phosphorus is good conductor of electricity while white α-black is non-conductor.

(iv) Structure : β-black phosphorus has multiple layered structure same as graphite.



β – black phosphorus

[Distance between two layers is found 3.68Å]

Note : Order of reactivity or stability of allotropic forms of phosphorus.

Stability $\boxed{\text{Black} - \text{P} > \text{Red} - \text{P} > \text{White} - \text{P}}$

Reactivity $\boxed{\text{Black} - \text{P} < \text{Red} - \text{P} < \text{White} - \text{P}}$

(B) Oxides of phosphorus

There are various oxides of phosphorus

Oxides	P_4O_6	P_4O_7	P_4O_8	P_4O_9	P_4O_{10}
Oxidation state	All P = +3	3P = +3 1P = +5	2P = +3 2P = +5	3P = +5 1P = +3	Al P = +5

Note : Stable phosphorus oxides and most commonly used oxides are P_4O_6 and P_4O_{10}.

1. Phosphorus trioxide (P_4O_6) :

It is dimer of P_2O_3.

Preparation :

It is prepared by burning white phosphorus in presence of air (limited)

$$P_4 + 3O_2 \longrightarrow P_4O_6$$

Properties :

(i) By heating :

In presence of air on heating its gives phosphorus penta oxide.

$$P_4O_6 + 2O_2 \longrightarrow P_4O_{10}$$

(ii) Reaction on with water :

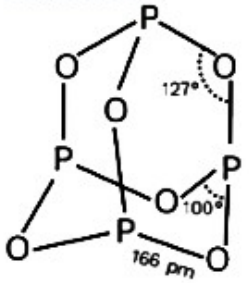
$$P_4O_6 + 6H_2O \text{ (Cold)} \longrightarrow 4H_3PO_3$$

phosphorus Acid

$$P_4O_6 + 6H_2O \text{ (Hot)} \longrightarrow 3H_3PO_4 + PH_3$$

Phosphoric acid

(iii) Structure :



Number of P–O–P bonds = 6

Number of P–P bond = 0

Number of lone pair = (6 × 2) + (4 × 1)

= 16

Hybridisation on P = sp^3

2. Phosphorous penta oxide (P_4O_{10}) :

Preparation :

It is prepared by burning white phosphorous in presence of air (excess)

$$P_4 + 5O_2 \longrightarrow P_4O_{10}$$

Properties :

(i) Reaction with acid :

$$2HNO_3 + P_4O_{10} \longrightarrow 2N_2O_5 + 4HPO_3$$

$$2H_2SO_4 + P_4O_{10} \longrightarrow 2SO_3 + 4HPO_3$$

In above reaction P_4O_{10} coverts acids into its anhydrous form and act as dehydrating agent

(ii) Reaction with water :

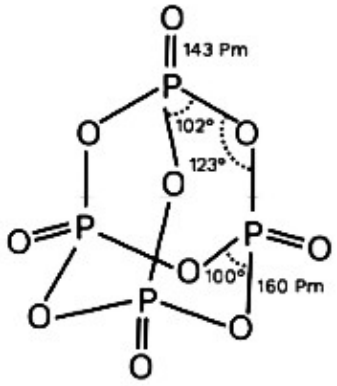
$$P_4O_{10} + 2H_2O\text{(cold)} \longrightarrow 4HPO_3$$

Meta phosphoric acid

$$P_4O_{10} + 6H_2O\text{(hot)} \longrightarrow 4H_3PO_4$$

Phosphoric acid

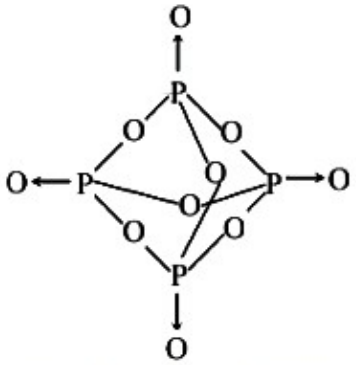
(iii) Structure :



Number of lone pairs = 20
Number of P–O–P bonds = 6
Number of $2P_{\pi}-3d_{\pi}$ bonds = 4
Hybridisation on P = sp^3

Q.31 The arrangement of oxygen atoms around phosphorus atoms in P_4O_{10} is -
(A) Pyramidal (B) Octahedral (C) Square planar (D) Tetrahedral

Sol.[D]



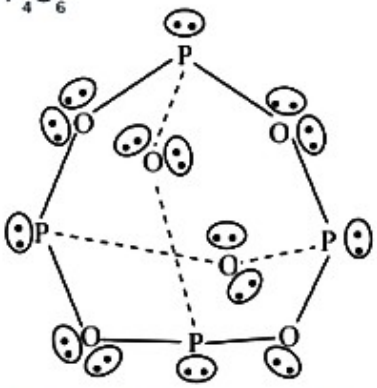
The arrangement of oxygen atoms around phosphorus atoms in P_4O_{10} is tetrahedral

Q.32 **In P_4O_6 the number of oxygen atoms bonded to each P atom is–**

(A) 1.5 **(B) 2** **(C) 3** **(D)4**

Sol. **[C]**

P_4O_6



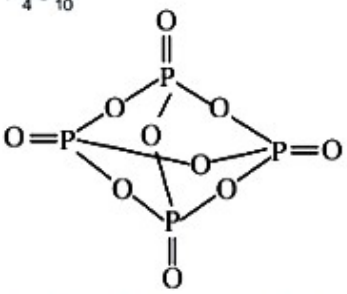
In P_4O_6 the number of oxygen atoms bonded to each P atom is three (3).

Q.33 **P_4O_{10} has short and long P–O bonds. The number of short P–O bonds in this compounds is–**

(A) 1 **(B) 2** **(C) 3** **(D) 4**

Sol. **[D]**

P_4O_{10}



Number of short P—O bonds in four.

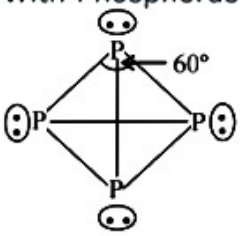
Q.34 **White phosphorus (P_4) has–** **[IIT -1998]**

(A) Six P–P single bonds **(B) Four P–P single bonds**

(C) Four lone pairs of electrons **(D) PPP angle of 60º**

Sol. **[A,C,D]**

With Phosphorus (P_4)



→ Six P-P single bonds
→ Four lone pairs of electrons
→ PPP angle is 60°

(C) Oxy Acids of Phosphorus

1. Phosphorus acid/ortho phosphorus acid [H_3PO_3]

Preparation :

(i) By reaction of P_4O_6 with water :

$$P_4O_6 + 6H_2O \longrightarrow 4H_3PO_3$$

(ii) By reaction of PCl_3 with water :

$$PCl_3 + 3H_2O \longrightarrow H_3PO_3 + 3HCl$$

Properties :

(i) It is a white crystalline solid.

(ii) It is soluble in water.

(iii) Its melting point is 74°C.

(iv) On neutralization, they form neutral salts called as phosphites.

$$H_3PO_3 + NaOH \longrightarrow NaH_2PO_3 + H_2O$$

$$NaH_2PO_3 + NaOH \longrightarrow Na_2HPO_3 + H_2O$$

(iv) Phosphorous acid also act, as strong reducing agent.

(a) Reduction of $AgNO_3$

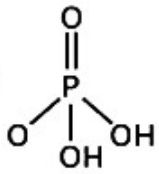
$$HPO_3^{2-} + 2Ag^+ \longrightarrow 2Ag\downarrow \text{ (black)} + H_3PO_4$$

(b) Mercury (II) chloride on heating with excess of the phosphite solution gives grey metallic mercury.

$$HPO_3^{2-} + 2HgCl_2 + H_2O \longrightarrow Hg_2Cl_2\downarrow \text{ (white)} + 2Cl^- + H_3PO_4$$

$$Hg_2Cl_2\downarrow \text{ (white)} + HPO_3^{2-} + H_2O \longrightarrow 2Hg\downarrow \text{ (grey)} + 2Cl^- + H_3PO_4$$

(v) It is dibasic acid.

Structure : 

2. Phosphoric acid /Orthophosphoric acid (H_3PO_4) :

Preparation :

(i) By Reaction of calcium phosphate with conc. H_2SO_4

$$Ca_3(PO_4)_2 + 3H_2SO_4 \longrightarrow 2CaSO_4 + 2H_3PO_4$$

(ii) On hydrolysis of PCl_5

$$PCl_5 + 4H_2O \longrightarrow H_3PO_4 + 5HCl$$

(iii) Reaction of white phosphorus in presence of concentrated HNO_3 :

$$P + 5HNO_3 \longrightarrow H_3PO_4 + 5NO_2 + H_2O$$

Properties :

(i) It is white crystalline solid.

(ii) It is soluble in water.

(iii) It contain weak acidic character.

(iv) Formation of Tollen's reagent

$$HPO_4^{2-} + 3Ag^+ \longrightarrow Ag_3PO_4\downarrow \text{ (yellow)} + H^+$$

$$Ag_3PO_4\downarrow\text{(yellow)} + 6NH_3 \longrightarrow 3[Ag(NH_3)_2]^{+1} + PO_4^{2-}$$

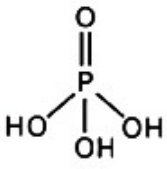
(v) Neutralization reaction :

$$H_3PO_4 \xrightarrow[-H_2O]{NaOH} NaH_2PO_4 \text{ (pri. phosphate)} \xrightarrow[-H_2O]{NaOH} Na_2HPO_4 \xrightarrow[-H_2O]{NaOH} Na_3PO_4 \text{ (tert. phosphate)}$$

(vi) Action of heat :

$$2H_3PO_4 \xrightarrow[-H_2O]{220°C} H_4P_2O_7 \text{ (pyrophosphoric acid);}$$

$$H_4P_2O_7 \xrightarrow{316°C} HPO_3 \text{ (metaphosphoric acid)}$$

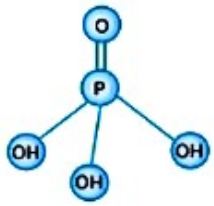
(vii) Structure of H_3PO_4 ⇒  (Tribasic acid)

Oxoacids of Phosphorus

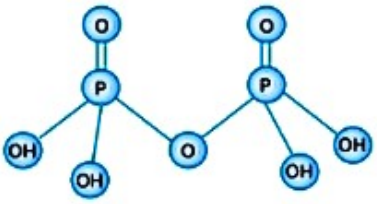
Name	Formula	Oxidation state of Phospho-rus	Characteristic bonds and their number	Preparation methods
Hypophospho-rus (Phosphinic)	H_3PO_2	+1	One P–OH Two P–H One P=O	white P_4 + alkali
Orthophos-phorous (Phosphnic)	H_3PO_3	+3	Two P–OH One P–H One P=O	$P_2O_3 + H_2O$
Pyrophospho-rus	$H_4P_2O_5$	+3	Two P–OH Two P=O Two P–H	$PCl_3 + H_3PO_3$
Hypophos-phoric	$H_4P_2O_6$	+4	Four P–OH Two P=O One P–O	red P_4 + alkali
Orthophos-phoric	H_3PO_4	+5	Three P–OH One P=O	$P_4O_{10} + H_2O$
Pyrophosphor-ic	$H_4P_2O_7$	+5	Four P–OH Two P=O One P–O–P	heat phos-phoric acid
Metaphos-phoric	$(HPO_3)_n$	+5	Three P–OH Three P = O Three P–O–P	Phosphorus acid + Br_2, heat in a sealed tube

Exists in polymeric forms only. Characteristic bonds of $(HPO_3)_3$ have been given in the table.

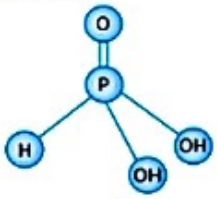
The chemical formulas and structures of various oxoacids.



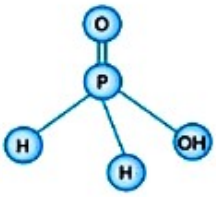
H_3PO_4
orthophosphoric acid



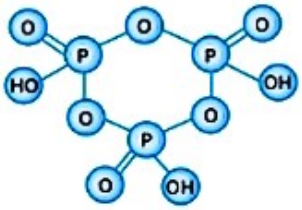
$H_4P_2O_7$
Pyrophosphoric acid



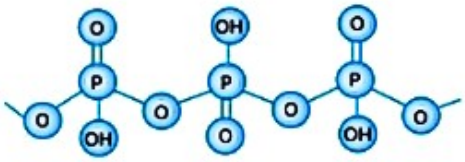
H_3PO_3
Orthophosphoric acid



H_3PO_2
Hypophosphoric acid



Cyclotrimetaphosphoric acid
$(HPO_3)_3$



Polymetaphosphoric acid
$(HPO_3)_n$

Q.35 **The number of P–O–P bonds in cyclic metaphosphoric acid is–**

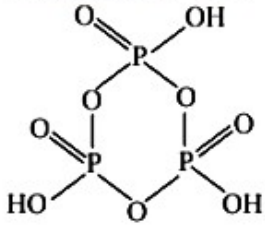
[IIT - 2000]

(A) Zero **(B) Two**

(C) Three **(D) Four**

Sol. **[C]**

Number of P–O–P bonds = 3 in cyclic trimetaphosphoric acid is there



Q.36 **Phosphorus pentoxide is used in the preparation of N_2O_5**

Sol. $2HNO_3 \rightarrow H_2O + N_2O_5$
$P_2O_5 + H_2O \rightarrow 2HPO_3$
$P_2O_5 + 2HNO_3 \rightarrow 2HPO_3 + N_2O_5$

Q.37 **White phosphorus contains -**

(A) P_2 molecules **(B) P_6 molecules**
(C) P_4 molecules **(D) P_5 molecules**

Sol. **[C]** white phosphorus contain molecules of P_4

Q.38 **By the action of conc. H_2SO_4, phosphorus changes to**

(A) H_3PO_3 **(B) HPO_3** **(C) H_3PO_4** **(D) $H_4P_2O_7$**

Sol. **[C]** $P_4 + 10\ H_2SO_4 \rightarrow 10SO_2 + 4H_3PO_4 + 4H_2O$

Important Compounds of Nitrogen Family

1. Ammonia (NH_3) :

Preparation :

(i) By decay of urea

$$NH_2CONH_2 + 2H_2O \longrightarrow (NH_4)_2CO_3 \rightleftharpoons 2NH_3 + H_2O + CO_2$$

(ii) Laboratory preparation :

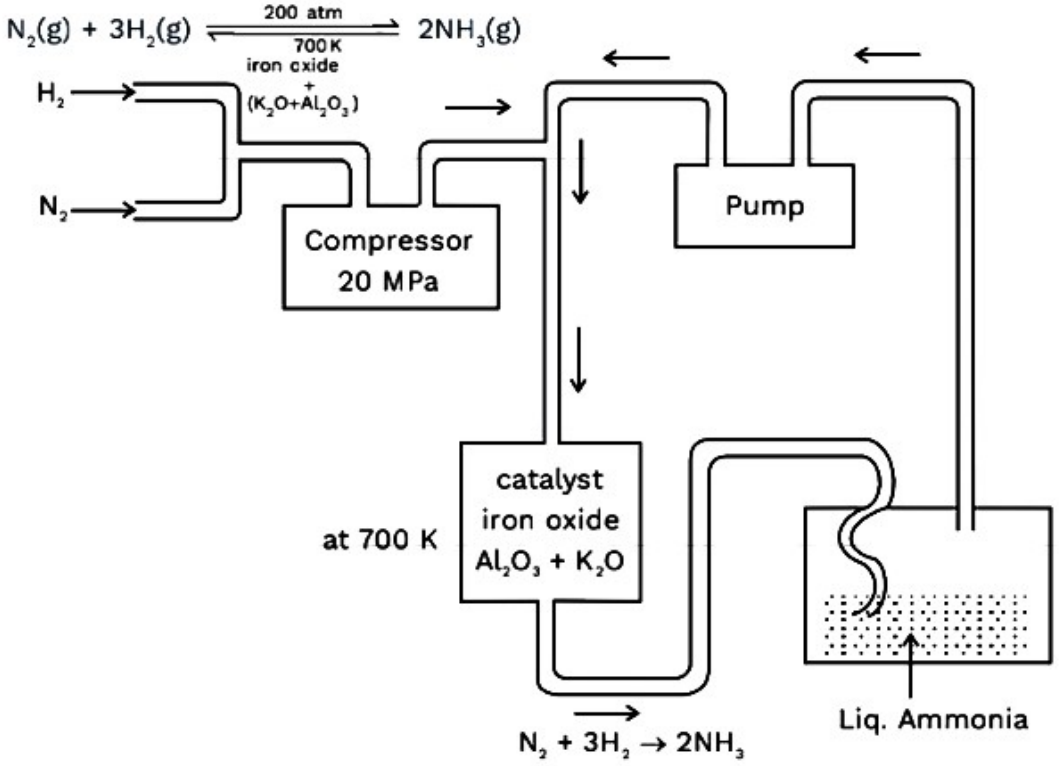
$$2NH_4Cl + Ca(OH)_2 \longrightarrow 2NH_3 + 2H_2O + CaCl_2$$

$$(NH_4)_2SO_4 + 2NaOH \longrightarrow 2NH_3 + 2H_2O + Na_2SO_4$$

(iii) Large scale (By Haber' Process)

$$N_2(g) + 3H_2(g) \rightleftharpoons 2NH_3(g)\ ;\ \Delta H = 46.1\ kJ/mole$$

Haber's Process Flow Charts :



Properties of Ammonia :

(i) It is colourless gas with pungent odour.

(ii) Freezing point = 198.4 K
Boiling point = 239.7 K

(iii) Its aqueous solution is basic and highly soluble in water.

$$NH_3(g) + H_2O(l) \rightleftharpoons \underbrace{NH_4^+(aq.) + OH^-(aq.)}_{NH_4OH}$$

(iv) It act as weak base and form precipitates the hydroxide of many metals halides.

$$2FeCl_3(aq.) + 3NH_4OH(aq.) \longrightarrow Fe_2O_3.xH_2O(s) + 3NH_4Cl(aq.)$$
(Brown ppt.)

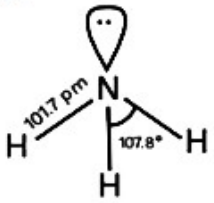
$$ZnSO_4(aq.) + 2NH_4OH(aq.) \longrightarrow Zn(OH)_2(s) + (NH_4)_2SO_4(aq.)$$
(white ppt)

(v) Forms complex compound with metal ions.

$$Cu^{+2} + 4NH_3 \longrightarrow [Cu(NH_3)_4]^{+2}$$

$$AgCl + 2NH_3 \longrightarrow [Ag(NH_3)_2]Cl$$

(vi) Structure :

$NH_3 \Rightarrow$ 

Bond angle = 107.8°
Bond length = 101.7 pm
Hybridisation = sp^3

Uses :

(i) In production of fertilizers
(ii) Used as refrigerant
(iii) Formation of some inorganic compound like nitric acid.

Q.39 Ammonium nitrate is formed in

(A) Zn + very dil. HNO_3 **(B) Fe + very dil. HNO_3**
(C) Sn + dil. HNO_3 **(D) All of these**

Sol. [D] $4Zn + 10\ HNO_3 \rightarrow 4Zn(NO_3)_2 + NH_4NO_3 + 3H_2O$
V. dil
$4Fe + 10HNO_3 \rightarrow 4Fe(NO_3)_2 + NH_4NO_3 + 3H_2O$
Very. dilute
$4Sn + 10HNO_3 \rightarrow 4Sn(NO_3)_2 + NH_4NO_3 + 3H_2O$

Q.40 Which of the following is /are correct for NH_3 and PH_3 ?

(A) Both are reducing agent. **(B) Both form complex compound**
(C) PH_3 is weaker base than NH_3 **(D) PH_3 is softer lewis base than NH_3**

Sol. [A,B,C,D]
(i) Both behave as reducing agent because both are present in its middle oxidation (+3) state thus it can further oxidises
(ii) Both form complex compounds
$2NH_3 + H_2PtCl_6 \rightarrow (NH_4)_2PtCl_6$
Ammonium chloroplatinate (yellow)
$AlCl_3 + 2PH_3 \rightarrow AlCl_3 . 2PH_3$
(iii) NH_3 more lewis base than PH_3
(iv) Due to large size of P it is soft lewis base

2. Posphine (PH_3) :

Preparation :

(i) By reaction of Ca_3P_2 with water or dilute HCl

$$Ca_3P_2 + 6H_2O \longrightarrow 3Ca(OH)_2 + 2PH_3$$

$$Ca_3P_2 + 6HCl \longrightarrow 3CaCl_2 + 2PH_3$$

(ii) In the laboratory: By heating white phosphorous with concentrated NaOH

$$P_4 + 3NaOH + 3H_2O \longrightarrow PH_3 + 3NaH_2PO_2$$

(sodium hypophosphite)

Properties :

(i) It is a colourless gas with rotten fish smell and it is highly poisonous

(ii) It is slightly soluble in water but soluble in CS_2.

(iii) It is weakly basic like ammonia, give phosphonium compound with acids

$$2PH_3 + HBr \longrightarrow PH_4^+Br^-$$ (phosphonium bromide)

Uses :

(i) It is used in Holme's signals.

(ii) Used in the manufacturing of smoke screens. It is due to formation of phosphorus pentaoxide.

Q.41 **PH_3 produces smoky rings when it comes in contact with air. This is because**

(A) it is inflammable **(B) it combines with water vapour**

(C) it combines with nitrogen **(D) it contains impurity of P_2H_4**

Sol. **[D]** Holme's signal work on principle that gaseous mixture of C_2H_2 and PH_3 catches five in presence of P_2H_4 spontaneously forming vortex rings white smoke and C_2H_2 produces a bright luminous flame

Q. 42 **White phosphorous (P_4) is boiled with a strong solution of sodium hydroxide in an inert atmosphere.** **[IIT-1987, 93]**

Sol. Phosphine gas (PH_3) is evolved when white phosphorus is boiled with aqueous NaOH or alcoholic solution of potassium hydroxide

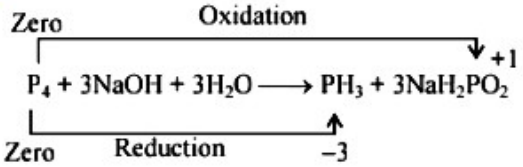
$$P_4 + 3NaOH + 3H_2O \rightarrow \underset{\text{Sodium hypo Phosphit}}{3NaH_2PO_2} + PH_3$$

Q.43 **White phosphorus on reaction with NaOH gives PH_3 as one of the products. This is a-**

(A) dimerization reaction **(B) disproportionation reaction**

(C) condensation reaction **(D) precipitation reaction**

Sol.



Some elements oxidised and reduced thus it is example of disproportion ation reaction.

Oxygen Family

Elements	Electronic configuration	Nature
$_{8}O^{16}$–oxygen	[He] $2s^2\ 2p^4$	non-metal
$_{16}S^{32}$–Sulphur	[Ne] $3s^2\ 3p^4$	non-metal
$_{54}Se^{78}$–Selenium	[Ar] $3d^{10}\ 4s^2\ 4P^4$	metalloids
$_{52}Te^{127}$-Tellurium	[Kr] $4d^{10}\ 5s^2\ 5p^4$	metalloid
Po–Polonium	[Xe] $4f^{14}\ 5d^{10}\ 6s^2\ 6p^4$	metal

Oxygen family also known as chalcogens.

Atomic and Physical Properties

1. **Atomic size (A.S.) :**

Atomic and ionic radius increases as going top to bottom in group.

	O	<	S	<	Se	<	Te	<	Po
Covalent radius (pm)	74		103		119		142		

⟶ Atomic size ↑

2. **Ionisation energy (I.E.) :**

Ionisation energy decreases as going down the group down the group due to increase in size. The value of ionisation energy is less compare then group-15 element due to half filled p-orbitals.

	O	>	S	>	Se	>	Te	>	Po
I.E. (kJ/mole)	1314		1000		941		869		

⟶ Ionisation Energy ↓

3. **Electronegativity (E.N.) :**

Electronegativity decrease as going down the group due to metallic character increases.

	O	>	S	>	Se	>	Te	>	Po
E.N.	3.5		2.44		2.48		2.01		

⟶ Electronegativity ↓

4. **Electron Affinity [E.A.] :**

Electron affinity decrease down the group except oxygen and sulphur, because 3rd period element has greater electron affinity compare then 2nd period element due to big size of orbital.

S > Se > Te > Po > O

⟶ Electron affinity ↓

5. **Physical state :**
Oxygen found in form of O_2 due to small size of oxygen so that oxygen form easily π bond : while S, Se and Te are found in S_8, Se_8 and Te_8 (Polyatomic) form respectively.

6. **Catenation property :**
Tendency for catenation decreases down the group. Prominently sulphur show catenation due to oxygen has small size and it has high repulsion between lone-pair so it has low bond energy.

$$\boxed{S > Se > O > Te}$$

Q.44 **Which bond have maximum bond energy :**
(A) C–C **(B) N–N** **(C) O–O** **(D) F–F**

Sol. **(A)**

7. **Allotropy :**
(A) oxygen has two allotropes O_2 and O_3 this type of allotropy is called as monotropy

Oxygen → O_2 (Oxygen gas)
Oxygen → O_3 (Ozone)

(B) Sulphur has number of allotropic forms.

Sulphur → (1) α-sulphur (Rhomibc-S)
Sulphur → (2) β-sulphur (monoclinic)
Sulphur → (3) γ-sulphur (plastic-S)
Sulphur → (4) milk of sulphur
Sulphur → (5) δ-sulphur (colloidal sulphur)

(C) Selenium has unique property of photo conductivity and is used in photocopying machine and also a decolorizer of glass.

8. **Oxidation state :**
→ Oxygen mostly slows -2 oxidation state because its electronegativity is high. Stability

of -2 oxidation state decrease down to the group.

→ Se, S and Te have vacant d-orbital so they show +4, +6 oxidation state also.

→ Stability of +6 oxidation state decrease down the group and stability of +4 oxidation state increase down the group due to inert pair effect.

→ Bonding in +4 and +6 oxidation states are primarily covalent in nature.

→ Exception : In OF_2 and O_2F_2 oxygen show +2 and +1 oxidation state respectively.

9. Density :

As going down the group density increases due to increase in mass.

Density	O	<	S	<	Se	<	Te
(gm/cm^3)	1.32		2.06		4.19		6.25

10. Melting and boiling point :

As going down the group the melting and boiling point increases

	O	<	S	<	Se	<	Te
M.P. (K)	54		393		490		725
B.P.(K)	90		718		958		1260

Q.45 Which shows maximum catenation property ?

(A) Te (B) Se (C) S (D) O

Sol. **[C]** Sulphur has highest power of catenation

O < S > Se > Te > Po

Q.46 The highest bond strength is shown by

(A) O – O bond (B) S – S bond

(C) Se – Se bond (D) Te – Te bond

Sol. **[B]** S – S Bond energy = 213 kJ/mol

O – O Bond energy = 138 kJ/mol

Q.47 **Which of following elements is highest electronegative ?**
(A) S (B) Se (C) Te (D) O

Sol. **[D]** O is highest electronegative element among S, Se, Te and O.

Q.48 **Arrange the following–**
(a) Po, Se, Te, S, O in order of increasing metallic properties.
(b) Po, Se, Te, S, O in order of decreasing ionisation potential.

Sol. (a) O < S < Se < Te < Po
Metallic properties increase
(b) O > S > Se > Te > Po
← Ionisation potential

Q.49 **The electron affinity of the members of oxygen of the periodic table, follows the sequence – [REE- 1990]**
(A) O > S > Se (B) S > O > Se (C) S > Se > O (D) Se > O > S

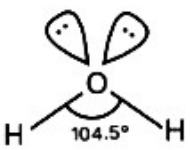
Sol. **[C]**
Electron affinity S > Se > O

Chemical Properties :

1. Hydrides :
All for elements of group-16 form hydrides like H_2O, H_2S, H_2Se, H_2Te, H_2Po

(i) Hybridisation :
In all hydrides central atom have sp^3 hybridisation, but centre atom have two lone-pairs so shape is bent or v-shape.

Example :



(ii) Thermal stability :
As going down to the group thermal stability decreases because bond length increases

$$\boxed{H_2O > H_2S > H_2Se > H_2Te}$$

$\longrightarrow$ Thermal stability ↓

(iii) Reducing Property :

Increases as going down the group because bond length of M–H bond increases from top to bottom.

$$\boxed{H_2O < H_2S < H_2Se < H_2Te}$$

$\longrightarrow$ Reduction ↑

(iv) Bond Angle :

As size of atom (center atom) increases repulsion of lone pair is decreases so bond angle decreases as going down the group.

104.5° 92.5° 91° 90°

$$\boxed{H_2O > H_2S > H_2Se > H_2Te}$$

$\longrightarrow$ Bond Angle ↓

(v) Dipole moment :

Dipole moment decrease as going down the group.

$$\boxed{H_2O > H_2S > H_2Se > H_2Te}$$

$\longrightarrow$ dipole moment ↓

(vi) Boiling point :

As going down the group boiling point is increases but exceptionally water boiling point is maximum.

→ H_2O has maximum boiling point due to H-Bonding

$$\boxed{H_2S < H_2Se < H_2Te < H_2O}$$

B.P.(K)	213	232	269	373
M.P.(K)	188	208	222	273

→ From H_2S to H_2Te Vander wall force of attraction increases so boiling point increases.

(vii) Acidic strength :

As going top to bottom in the group M–H bond length increases so tendency to loose H^+ ion increases and Acid strength increases.

$$\boxed{H_2O < H_2S < H_2Se < H_2Te}$$

$\longrightarrow$ Acid strength ↑

(viii) Covalent character :

As going top to bottom in group, as electronegativity of atom decreases, tendency to share electron increases. So covalent character increases from top to bottom.

$$\boxed{H_2O < H_2S < H_2Se < H_2Te}$$

$\longrightarrow$ Covalent character ↑

(xi) Reactivity with hydrogen :

As going down to group hydrides reactivity increases due to decrease bond. dissoci-

ation energy of H–E bond. (E = Element)

Reactivity ⇒	H_2O <	H_2S <	H_2Se <	H_2Te
Bond dissociation ⇒ Energy (kJ/mole)	463	347	276	238

Q.50 **H_2S oxidises in presence of excess oxygen, gives -**

(A) SO_2 **(B) SO_3** **(C) S** **(D) H_2SO_4**

Sol. **[A]** $2H_2S + 3O_2 \rightarrow 2H_2O + 2SO_2$

Q.51 **Oxygen is more electronegative than sulphur yet H_2S is more acidic than H_2O. Because –**

(A) stronger inter molecular H-bonding among H_2O

(B) H_2O have higher boiling point than H_2S

(C) H-S is weaker than O – H bond

(D) H_2S is a gas at ordinary temperature where as H_2O is a liquid

Sol. **[C]** H_2S is more acidic than H_2O because H–S bond is weaker than O – H bond

Q.52 **Arrange the following–**

(a) H_2S, H_2O, H_2Se, H_2Te in order of increasing acidity.

(b) H_2O, H_2S, H_2Se, H_2Te in order of increasing thermal stability.

(c) H_2O, CO_2, SO_3, N_2O_5, SiO_2 in order of decreasing acidity.

Sol. (a) $H_2O < H_2S < H_2Se < H_2Te$
Acidity increases

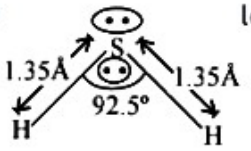
(b) $H_2O > H_2S > H_2Se > H_2Te$
← Thermal Stability

(c) $SO_3 > N_2O_5 > CO_2 > SiO_2 > H_2O$
← Acidity

Q.53 **In H_2O molecule what is co-ordination number of O ?**
(A) 6 **(B) 2** **(C) 4** **(D) zero**

Sol. **[C]**
In H_2O molecules coordination number of oxygen is four.

Q.54 **The geometry of H_2S and its dipole moment are –** **[IIT - 1999]**
(A) angular and non-zero **(B) angular and zero**
(C) linear and non-zero **(D) linear and zero**

Sol. **[A]**
H_2S molecules has V-shaped structure. With bond length (H–S) 1.35 Å and bond angle (H–S–H) 92.5º 

lone pair of electron Centre atom has lone pair of electron thus its dipole moment is never zero ($\mu \neq 0$).

2. Oxides :
Group-16 elements form mono oxides, dioxides, and trioxides.

Element	Monooxide	Dioxide	Trioxide
Sulphur	SO	SO_2	SO_3
Selenium	–	SeO_2	SeO_3
Tellurium	TeO	TeO_2	TeO_3
Polonium	PoO	PoO_2	–

→ Acidic strength of oxides increases as oxidation state on central atom increases.

$$\overset{+2}{SO} < \overset{+4}{SO_2} < \overset{+6}{SO_3}$$

⟶ Acid strength ↑

→ If oxidation state is same on central atom then acidic strength decreases as electro-negativity of central atom decreases.

$$\overset{+4}{SO_2} > \overset{+4}{SeO_2} > \overset{+4}{TeO_2} > \overset{+4}{PoO_2}$$

⟶ Acidic strength ↓

Q.55 **Ozone acts as**

(A) Oxidising agent **(B) reducing agent**

(C) bleaching agent **(D) all**

Sol. **[D]** ozone acts as

$O_3 \rightarrow O_2 + [O]$ oxidising agent & bleaching agent $O_3 + (O) \rightarrow 2O_2$ reducing agent

Q.56 **Which of the following oxides is a basic oxide**

(A) PbO **(B) SiO_2** **(C) SnO_2** **(D) CrO_3**

Sol. **[A]** PbO is amphoteric oxide

$PbO + 2HNO_3 \rightarrow Pb(NO_3)_2 + H_2O$

$PbO + 2NaOH \rightarrow Na_2PbO_2 + H_2O$

Q.57 **Oxide of a non-metal possesses the following characteristics–**

(i) **It is both a proton donor and an acceptor**

(ii) **It is a poor conductor of electricity**

(iii) **It reacts readily with basic and acidic oxides**

(iv) **It oxidises Fe at its boiling point This oxide is**

(A) SO_2 **(B) CO_2** **(C) H_2O_2** **(D) H_2O**

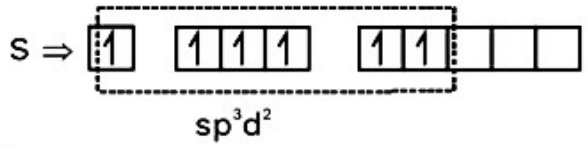
Sol. **[D]**
Oxide of non metal possesses poor conductor of electricity.

Q.58 **SO_2 is more powerful reducing agent in–**
(A) Acidic medium **(B) Alkaline medium**
(C) Neutral medium
(D) Equally powerful reducing agent in acidic and alkaline solution

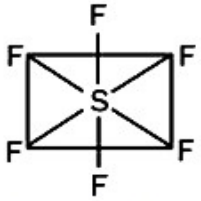
Sol. **[B]**

3. **Halides :**
Group-16 elements form large number of halides. The stabilities of the halides decreases in order F > Cl > Br > I

(i) **Hexahalide :**
Hybridisation ⇒ sp^3d^2
shape ⇒ octahedral
Example : SF_6



sp^3d^2



Octahedral

Note : (1) Sulphur has vacant d-orbitals, but it cannot be hydrolysis because six small size F. (It is called as steric protection)

order hydrolysis $\boxed{SF_6 < SeF_6 < TeF_6}$

order of stability $\boxed{SF_6 > SeF_6 > TeF_6}$

Size ↑, Stability ↓

(2) Amongst hexahalides, hexaflourideds are only stable halides and gaseous in nature.

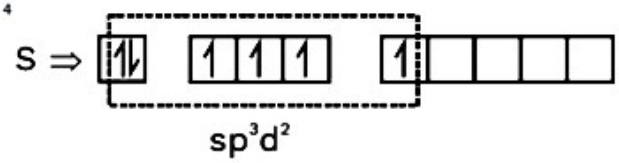
(ii) Tetra halide :

Hybridisation = sp^3d

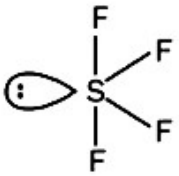
Shape = see saw (due to lone pairs)

(Trigonal bipyramidal)

Example : SF_4



sp^3d^2

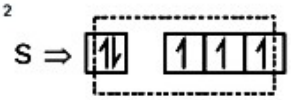


(iii) Di halide :

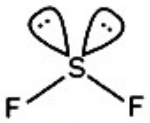
Hybridisation = sp^3

Shape = bent or V-shape

Example : SF_2



sp^3



Note : Except selenium all elements form dichloride and dibromides.

(iv) Monohalide

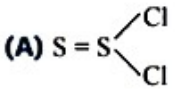
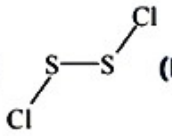
The well known monohalide are dimeric in nature.

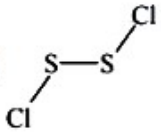
Example : S_2F_2, S_2Cl_2, S_2Br_2, Se_2Br_2

Dimeric halides undergo disproportionation

$2Se_2Cl_2 \longrightarrow SeCl_4 + 3Se.$

Q.59 **Which of the following is the correct structure of S_2Cl_2 ?**

(A)  **(B) Cl – S – S – Cl** **(C)**  **(D) Cl – S = S – Cl**

Sol. **[C]** Correct structure of S_2Cl 

Q.60 **SF_6 not hydrolysed while SF_4 is hydrolysed**

Sol. **True**

SF_6 is not hydrolysed because it is completely bonded.

Compound of oxygen and their oxides

1. **Dioxygen [O_2] :**

Preparation :

(i) By heating potassium chlorate

$$2KClO_3 \longrightarrow 2KCl + 3O_2$$

(ii) Thermal decomposition of oxides

Metals which are low reactive, their oxides give oxygen gas on their thermal decom position.

$$2Ag_2O \xrightarrow{\Delta} 4Ag + O_2$$

$$2HgO \xrightarrow{\Delta} 2Hg + O_2$$

(iii) Industrially it is obtaining from air.

Properties :

(i) It is colouless and ordourless gas

(ii) It has three isotops O^{16}, O^{17} and O^{18}.

(iii) According to molecular orbital theory O_2 is paramagnetic in nature

(iv) It reacts with most of metals and non-metals

From metals $\begin{bmatrix} 2Ca + O_2 \longrightarrow 2CaO \\ 4Al + 3O_2 \longrightarrow 2Al_2O_3 \end{bmatrix}$

From non-metal $\begin{bmatrix} P_4 + SO_2 \longrightarrow P_4O_{10} \\ C + O_2 \longrightarrow CO_2 \end{bmatrix}$

From some compounds $\begin{bmatrix} 2ZnS + 3O_2 \longrightarrow 2ZnO + 2SO_2 \\ CH_4 + 2O_2 \longrightarrow CO_2 + 2H_2O \end{bmatrix}$

Uses :

(i) Use in respiration and combustion

(ii) Used in welding process.

Q.61 **In the electrolysis method of acidified water to give O_2, the cathode used is**

(A) graphite **(B) lead** **(C) platinum** **(D) Nickel**

Sol. **[C]** In the electrolysis method of acidified water to give O_2, the cathode is used in platinum

Q.62 **Freezing point of O_2 is -**

(A) – 183ºC **(B) – 229ºC** **(C) – 195.8ºC** **(C) – 186ºC**

Sol. **[B]** Freezing point of O_2 is –229° C

Q.63 **The number of unpaired electrons in a paramagnetic diatomic molecule of an element with atomic number 16 is –**

(A) 4 **(B) 1** **(C) 2** **(D) 3**

Sol. **[C]**

Q.64 **Oxygen is denser than air so it is collected over–**

(A) H_2O **(B) Spiri** **(C) Mercury** **(D) Kerosene oil**

Sol. **[A]**

Oxygen is denser than air so it is collected over H_2O.

Q.65 **Oxygen has + 2 oxidation state in the compound –**

(A) MgO **(B) F_2O** **(C) Na_2O** **(D) H_2O**

Sol. **[B]**

$F_2O = 2(-1) + x = 0$

$x = +2$

Q.66 **Which of the following compound on heating produces $O_2(g)$–**

(A) $KClO_4$ **(B) $NaNO_3$** **(C) $KMnO_4$** **(D) NH_4NO_2**

Sol. **[A,B,C]**

$KClO_4 \xrightarrow{\Delta} KCl + 2O_2$

$2NaNO_3 \xrightarrow{\Delta} Na_2NO_3 + NO_2 + 1/2O_2$

$KMnO_4 \xrightarrow{\Delta} K_2MnO_4 + MnO_2 + O_2$

$NH_4NO_2 \rightarrow N_2 + 2H_2O$

2. Ozone (O_3) :

Van marum found that, when electricity is flow in O_2 gas smell like rotten fish comes out. It is due to a new gas called ozone.

Preparation :

$O_2 \xrightarrow{\text{Electricity}} [O] + [O]$

$O_2 + [O] \longrightarrow O_3$; ΔH = High.

$3O_2 \rightleftharpoons 2O_3$

Note :

- Formation of ozone is endothermic process.
- O_2 and O_3 are found in equilibrium mixture.

  $90 - 95\% \Rightarrow O_2$

  $5 - 10\% \Rightarrow O_3$

This mixture is called ozonised oxygen.

Properties :

(1) Pure ozone is pale blue colour gas, Dark blue in liquid state and violet black in solid state

(2) It has characteristic smell like rotten fish. In small quantity it is harmless but high amount is cause of headache and nausea.

(3) It act as strong oxidizing agent due to nascent oxygen.

Example :

(i) Lead sulphide to lead sulphate

$$PbS_{(s)} + 4O_{3(g)} \longrightarrow PbSO_{4(s)} + 4O_2$$

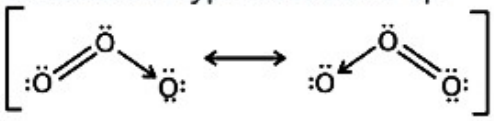
(Black) (White)

(ii) Iodide into iodine

$$2KI + H_2O + O_{3(g)} \longrightarrow 2KOH + I_{2(s)} + O_{2(g)}$$

Note : When ozone react with excess of KI and buffered with borate buffer (PH = 9.29) I_2 liberate which is treated with standard solution of sodium thio sulphate. This is quantitative method for estimating O_3 gas.

(5) **Structure :** Hypridisation $\Rightarrow sp^2$



Vapour phase

6. Nitrogen oxide emitted from supersonic jet slowly depleting the ozone layer.

$$NO(g) + O_3(g) \longrightarrow NO_2(g) + O_2(g)$$

7. Bleaching property :
It gives nascent oxygen, which converts colourful compound into colourless things.

Q.67 **Ozone on reacting with KI in neutral medium produces**

(A) KIO_3 **(B) KOH** **(C) KCl** **(D) KO_2**

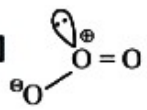
Sol. **[B]** $O_3 + 2KI + H_2O \rightarrow 2KOH + I_2 + O_2$

Q.68 **The structure of O_3 and N_3^- are –**

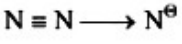
(A) Linear and bent respectively **(B) both linear**

(C) both bent **(D) bent and linear respectively**

Sol. **[D]**



bent



linear

Q.69 **Ozone is more reactive than oxygen because–**

(A) ozone is formed endothermically from oxygen.

(B) ozone possess higher energy.

(C) ozone on decomposition yields atomic oxygen

(D) ozone is triatomic

Sol. **[A,B,C]**

$$2O_3 \xrightarrow[300°C]{\text{heat}} 3O_2 + 68 \text{ k cal}$$

$$\downarrow$$

$$O_2 + O$$

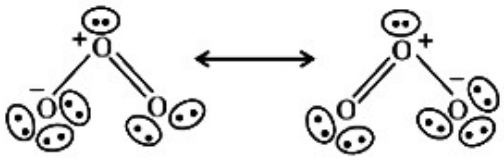
$\Rightarrow O_2 + O_2 \rightarrow O_3$

Ozone is formed endothermically.

Q.70 **Write two resonance structures of ozone which satisfy the octet rule.**

[IIT - 1991]

Sol. The two resonance structure of ozone are



Sulphur and its oxy acids :

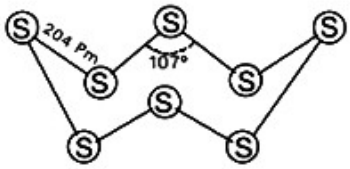
(A) Sulphur has different allotropic form

(i) α-sulpher/Rhombic sulphur :

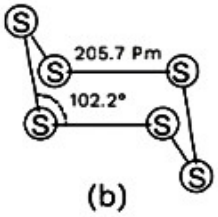
- It is most stable allotropic form of sulphur.
- When it is heated more than 95.5°C, it convert into β-sulphur.
- It is soluble in CS_2.

(ii) β-Sulphur/monoclinic :

- When α-sulphur is heated above 95.5°C then β-sulphur is obtained.
- It is less stable then α-sulphur
- It is also soluble in CS_2



(a)
S_8
Ring rhombic sulphur



(b)
S_6
Cyclo-S_6

(iii) γ-Sulphur/Plastic Sulphur :

- When boiled sulphur suddenly poured in cold water then γ-sulphur or plastic sulphur is obtained.
- It is very unstable
- It is not soluble in CS_2.

(iv) Milk of sulphur :

- It is formed by these following steps :

Step – I

$$\underset{\text{Milk of lime}}{Ca(OH)_2} + S \longrightarrow CaS_2 + \underset{\substack{\text{Calciumthio}\\ \text{sulphate}}}{CaS_2O_3} + H_2O$$

Step-II

$$CaS_5 + CaS_2O_3 + HCl \longrightarrow CaCl_2 + H_2O + \underset{\text{Milk of S}}{S}$$

- It is soluble in CS_2

(v) δ-Sulphur/Colloidal sulphur :

- When H_2S is treated with concentrated HNO_3 then δ-sulphur is obtained.
- It is soluble in CS_2

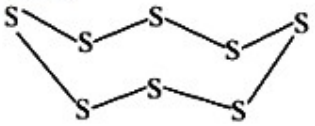
Q.71 **The number of lone pairs and the number of S-S bonds in S_8 molecules are respectively –**

(A) 8, 8 **(B) 16, 8** **(C) 8, 16** **(D) 8, 4**

Sol. **[B]** In S_8 molecule each atom is attached to two sulphur atom by singled bond and each sulphur has two lone pair, s-s bonds = 8, lone pair = 16

Q.72 **S_8 ring is non-planar –**

Sol. **True**



it is non planar molecules

Q.73 **Which of the following is not the application of sulphur–**

(A) in the vulcanisation of rubber **(B) as a antiseptic**

(C) in match industry **(D) alloying agent**

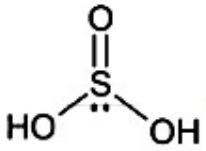
Sol. **[D]**

Sulphur doesn't participate in alloy formation.

Oxyacid of sulphur

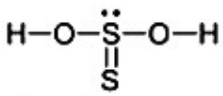
Sulphur forms number of oxoacids such as H_2SO_3, $H_2S_2O_4$, $H_2S_2O_5$, $H_2S_2O_6$, H_2SO_4, $H_2S_2O_7$, H_2SO_8, some of these acid are unstable and cannot be isolated.

1. **Sulphurous acid (H_2SO_3)**
   Structure :



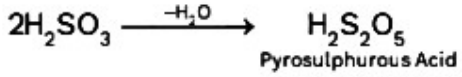
Hybridisation = sp^3

2. **Thio sulphurous Acid ($H_2S_2O_2$)**
   Structure :



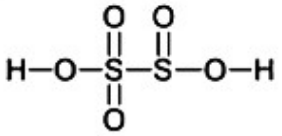
Hybridisation = sp^3

3. **Pyro sulphurous acid ($H_2S_2O_5$)**



Pyrosulphurous Acid

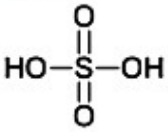
Structure :



Hybridisation = sp^3(both S)

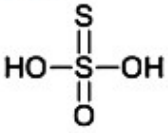
4. **Sulphuric acid (H_2SO_4)**

Structure :



5. **Thio sulphuric acid ($H_2S_2O_3$)**

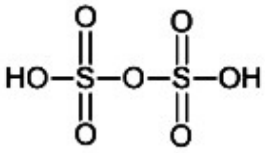
Structure :



6. **Pyrosulphuric acid ($H_2S_2O_7$)/oleum**

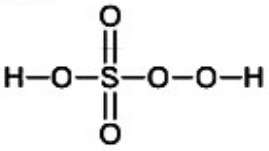
$2H_2SO_4 \xrightarrow{-H_2O} H_2S_2O_7$

Structure :



7. **Peroxy Acid/mono sulphuric acid (H_2SO_5)**

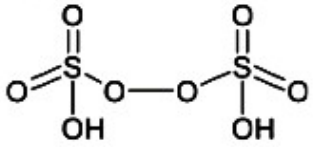
Structure :



It is also known as caro's acid.

8. **Peroxodisulphuric acid ($H_2S_2O_8$)**

Structure :



Some important compounds of Group-16

1. Sulphuric acid (H_2SO_4) :

(i) Lead Chamber Process :

In this method no catalyst used.

$$2SO_2 + O_2(air) + 2H_2O \xrightarrow[\text{catalyst}]{[NO]} 2H_2SO_4 + [NO]$$

- Acid obtained is 80% pure and is known as brown oil of vitriol.

(ii) Contact process :

Contact process involve following steps.

Step-I : Burning of Sulphur or Sulphide ore in air to generate SO_2.

$$S + O_2 \longrightarrow SO_2(g)$$

Step-II : Oxidation of SO_2 into SO_3 in the presence of catalyst V_2O_5.

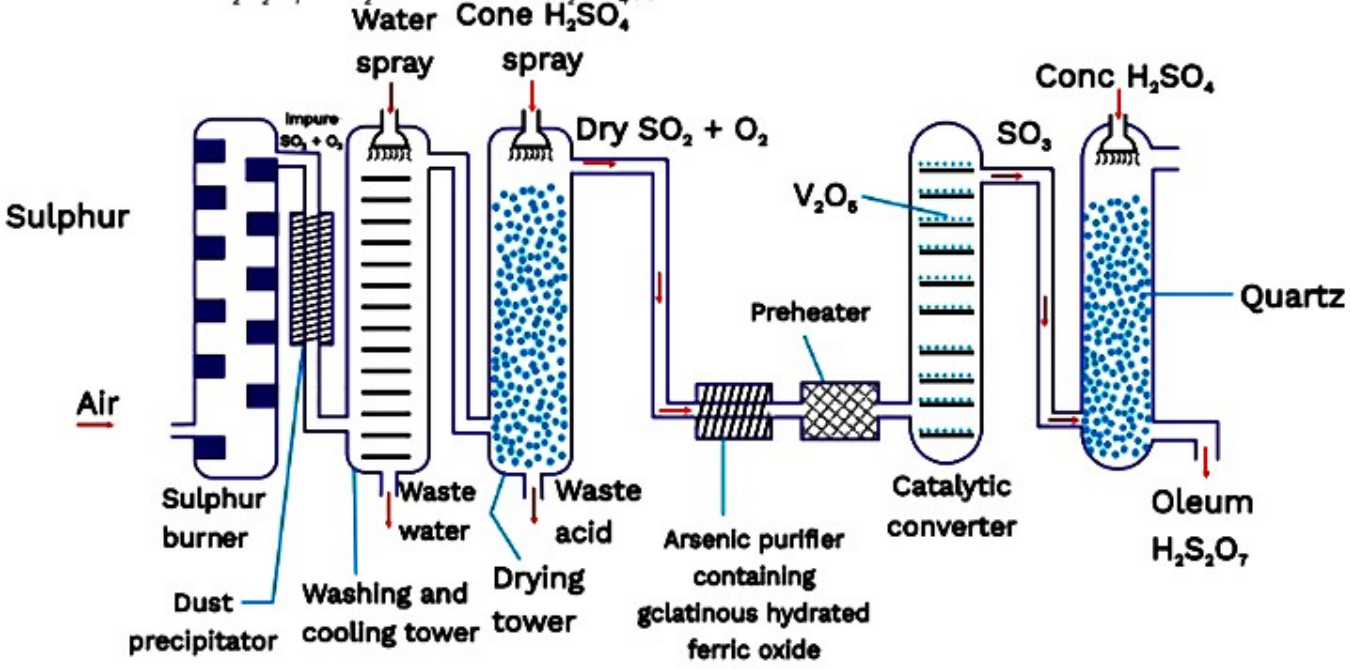
$$SO_2(g) + O_2(g) \xrightarrow{V_2O_5} 2SO_3(g) + \text{heat}$$

Step-III : Absorption of SO_3 in H_2SO_4 to give oleum ($H_2S_2O_7$)

$$H_2SO_4 + SO_3(g) \longrightarrow \underset{\text{(Oleum)}}{H_2S_2O_7}$$

Step-IV : Hydrolysis of oleum

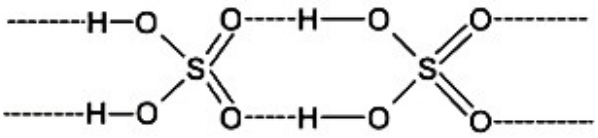
$$H_2S_2O_7 + H_2O \longrightarrow 2H_2SO_4(l)$$



Flow diagram for manufacturing of H_2SO_4

Properties :

1. It is a colourless liquid, its boiling point is high due to intra molecular H-bonding.



2. Acidic nature : It is a strong di-basic acid. It ionise in two steps

$H_2SO_4 \longrightarrow H^+ + HSO_4^-$
bisulphate ion

$HSO_4^- \longrightarrow H^+ + SO_4^{-2}$
Sulphate ion

3. It form two type of salts.

$H_2SO_4 + NaOH \longrightarrow NaHSO_4 + H_2O$
Sodium bisulphate
(Acidic salt)

$H_2SO_4 + 2NaOH \longrightarrow N_2SO_4 + 2H_2O$
Sodium sulphate
(simple salt)

4. It decompose carbonate, bicarbonate and give CO_2 gas :

$Na_2CO_3 + H_2SO_4 \longrightarrow Na_2SO_4 + H_2O + CO_2$

$NaHCO_3 + H_2SO_4 \longrightarrow NaHSO_4 + H_2O + CO_2$

5. It is a strong oxidising agent because it give nascent oxygen fast. Its oxiding properties can be understand by following examples.

$H_2SO_4 \longrightarrow H_2O + SO_2 + [O]$

Examples :

(i) It oxidies non-metals like carbon and Sulphur in their oxide.

$C + H_2SO_4 \longrightarrow CO_2 + SO_2 + H_2O$
(Conc.)

$S + H_2SO_4 \longrightarrow SO_2 + H_2O$
(Conc.)

(ii) It oxides P in phosphonic acid.

$P + H_2SO_4 \longrightarrow H_3PO_4 + SO_2 + H_2O$
(Conc.)

(iii) It oxides melts like Cu, Ag, Hg, first in their and after that convert them in sulphates.

$H_2SO_4 \longrightarrow H_2O + SO_2\,[O]$ ...(1)

$Cu + [O] \longrightarrow CuO$ ...(2)

$$CuO + H_2SO_4 \longrightarrow CuSO_4 + H_2O \quad ...(3)$$

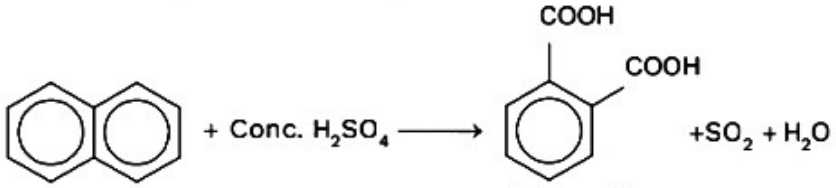
$$Cu + 2H_2SO_4 \longrightarrow CuSO_4 + SO_2 + 2H_2O$$

Similarly :

$$2Ag + 2H_2SO_4 \longrightarrow Ag_2SO_4 + SO_2 + 2H_2O$$

$$Hg + 2H_2SO_4 \longrightarrow HgSO_4 + SO_2 + 2H_2O$$

(iv) It oxidise naphthalene into pthalic acid.



Pthalic acid

6. **Dehydrating property :** It is a strong dehydrating agent. It removes molecules of H_2O in form of hydrate :
$H_2SO_4.H_2O$, $H_2SO_4.2H_2O$, $H_2SO_4.3H_2$ etc.

Example :

(i) $C_{12}H_{22}O_{11} \xrightarrow{\text{conc. } H_2SO_4} 12C + 11H_2O$
Sucrose

(ii) $C_6H_{12}O_6 \xrightarrow{\text{conc. } H_2SO_4} 6C + 6H_2O$
glucose

(iii) $\begin{matrix}COOH\\|\\COOH\end{matrix} \xrightarrow{\text{conc. } H_2SO_4} CO + CO_2 + H_2O$
Oxalic acid

(iv) $HCOOH \xrightarrow{\text{conc. } H_2SO_4} CO + H_2O$
Formic acid

Uses :

(i) Used in manufacturing of fertilizers
(ii) Petroleum refining
(iii) Detergent industry
(iv) Use in metallurgical process
(v) Use in lead storage batteries.
(vi) Used as laboratory agent.

Q.74 **H_2SO_4 has very high corrosive action on skin because–**

(A) it reacts with proteins

(B) It acts as an oxidising agent

(C) it acts as a dehydrating agent

(D) it acts as dehydrating agent and absorption of water is highly exothermic

Sol. **[D]**

It acts as dehydrating agent and absorption of water is highly exothermic.

Q.75 **Conc. H_2SO_4 displaces hydrogen chloride from chlorides because–**

(A) it is stronger than hydrochloric acid

(B) HCl is a gas while H_2SO_4 is a liquid

(C) sulphates are more soluble than chlorides

(D) sulphates are less soluble than chlorides

Sol. **[B]**.

Q.76 **In sulphate ion the oxidation state of sulphur is + 6 and hybridization state of sulphur is–**

(A) sp^2 **(B) sp^3d^2** **(C) dsp^3** **(D) sp^3**

Sol. **[D]**

SO_4^{2-}

$x + (-8) = -2$

$x = +6$

$SO_4^{2-} = sp^3$

Q.77 **Which of the following is the most powerful oxidising agent ?**

(A) H_2SO_4 **(B) H_3BO_3** **(C) HPO_3** **(D) H_3PO_4**

Sol. **[A]**

H_2SO_4 is most powerful oxidising agent because sulphur is present in its highest ox-

idation self (oxidises others self reduces).

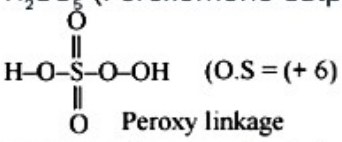
H_2SO_4	H_3BO_3	H_3PO_4
$2 + x(-8) = 0$	$+3 + x + (-6) = 0$	$+3+x+(-8) = 0$
$x = +6$	$x = +3$	$x = +5$

Q.78 **The peroxy linkage present in–**

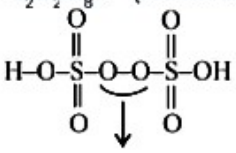
(A) H_2SO_5 **(B) $H_2S_2O_7$** **(C) $H_2S_2O_8$** **(D) $H_2S_2O_3$**

Sol. **[A,C]**

H_2SO_5 (Peroxomono sulphuric acid)

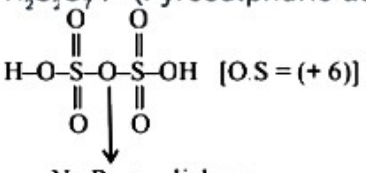


$H_2S_2O_8$:- (Peroxodisulphuric acid)



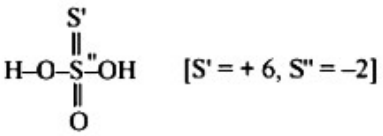
Peroxy linkage [O.S = (+ 6)]

$H_2S_2O_7$:- (Pyrosulphuric acid)



No Peroxy linkage

$H_2S_2O_3$ (Thio sulphuric acid)

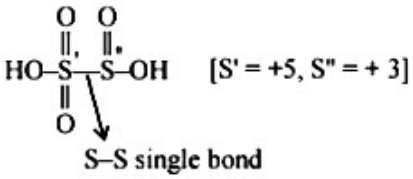


Q.79 **Which of the following oxyanions contain S–S (single bond)–**

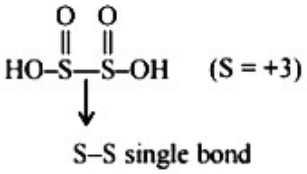
(A) $S_2O_3^{2-}$ **(B) $S_2O_4^{2-}$** **(C) $S_2O_5^{2-}$** **(D) $S_2O_7^{2-}$**

Sol. **[B,C]**

$H_2S_2O_5$:- (Pyrosulphurous acid)



$H_2S_2O_4$ (dithionous acid)



Q.80 **Ionically the reaction between H_2SO_4 & Mg may be presented as, $Mg + 2H_3O^+ \rightarrow Mg^{2+} + H_2 \uparrow + 2H_2O$. Therefore, in the given reaction–**

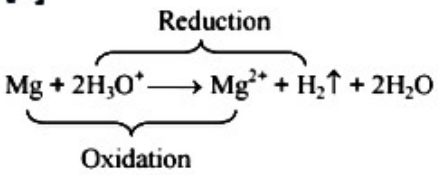
(A) oxidising agent is $H^+(H_2O)$

(B) reducing agent is $H^+(H_2O)$

(C) oxidising agent is SO_4^{2-}

(D) oxidising agents are SO_4^{2-} & H_3O^+ both

Sol. **[A]**



In this reaction H_3O^+ behave as oxidising agent (other oxidised self reduces)

Q.81 **$H_2SO_4 + PCl_5 \longrightarrow (X) \xrightarrow{H_2O}$ two strong acids Therefore (x) is –**

(A) SO_2Cl_2 **(B) SOCl** **(C) $ClSO_3H$** **(D) $POCl_3$**

Sol. **[A]**

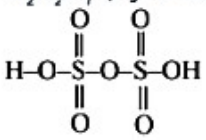
$$H_2SO_4 + PCl_5 \rightarrow \underset{SO_2Cl_2}{(X)} \xrightarrow{H_2O} SO_3 + 2HCl$$

Q.82 **There is no S–S bond in –** **[IIT - 1991]**

(A) $S_2O_4^{2-}$ **(B)** $S_2O_5^{2-}$ **(C)** $S_2O_3^{2-}$ **(D)** $S_2O_7^{2-}$

Sol. **[D]**

$H_2S_2O_7$ (Pyro sulphuric acid)



There is no S–S Bond

2. Sulphur dioxide (SO_2)

Preparation :

(i) By burning S in air

$$S(s) + O_2(g) \longrightarrow SO_2(g)$$

(ii) Laboratory method : By treating sulphite with dilute sulpuric acid

$$Na_2SO_3 + H_2SO_4 \longrightarrow Na_2SO_4 + H_2O + \boxed{SO_2(g)}$$

(iii) Industrially : It is prepared by roasting of sulphide ore

$$4FeS_2(s) + 11O_2(g) \longrightarrow 2Fe_2O_3(s) + 8SO_2(g)$$

Properties :

(i) Clourless gas with pungent smell and is highly soluble in water.

(ii) It is acidic in nature, it gives sulphurous acid with water :

$$H_2O + SO_2 \longrightarrow H_2SO_3$$

(iii) SO_2 react with Cl_2 in presence of charcoal to give sulphuryl chloride (SO_2Cl_2)

$$SO_2(g) + Cl_2(g) \longrightarrow SO_2Cl_2(g)$$

Sulphuryl chloride

(iv) It oxidise in SO_3 in presence of V_2O_5

$$SO_2(g) + O_2(g) \longrightarrow SO_3(g)$$

(v) It readily react with NaOH solution, forming sodium sulphite, which them react with more SO_2 form sodium hydrogen sulphite.

$$2NaOH + SO_2 \longrightarrow Na_2SO_3 + H_2O$$

$$Na_2SO_3 + H_2O + SO_2 \longrightarrow 2NaHSO_3$$

(vi) Bleaching property : In presence of water, it act as bleaching agent bleaching by SO_2 is unstable, because it does bleaching by nascent H. (Colourless compound again react with O_2 in air)

$$SO_2 + H_2O \longrightarrow H_2SO_3$$

$H_2SO_3 + H_2O \longrightarrow H_2SO_4 + Z[H]$

Note : Bleaching done by Cl_2 and O_3 is stable because they do bleaching by nascent [O]

(vii) SO_2 act as reducing as well as oxidising agent.

(A) Reducing property : In presence of H_2O it act as reducing agent

$SO_2 + H_2O \longrightarrow H_2SO_3$

$H_2SO_3 + H_2O \longrightarrow H_2SO_4 + [H]$

Example :

(i) It reduce $KMnO_4$

$$\overset{+7}{K}\overset{+7}{Mn}O_4 + SO_2 + H_2O \longrightarrow K_2SO_4 + \overset{+2}{Mn}SO_4 + H_2SO_4$$

(ii) It reduce $K_2Cr_2O_7$ in chromic sulphate $Cr_2(SO_4)_3$

$$K_2\overset{+7}{Cr_2}O_7 + H_2SO_4 + SO_2 \longrightarrow K_2SO_4 + Cr_2(SO_4)_3 + H_2O$$

green

(B) Oxidising property :

Example :

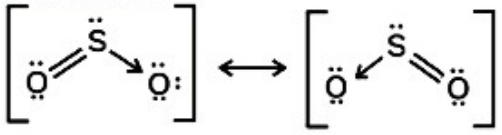
(i) Oxidise H_2S in S–

$H_2S + SO_2 \longrightarrow S + H_2O$

(ii) CO in CO_2

$CO + SO_2 \longrightarrow CO_2 + S$

(viii) Structure :



hybridisation $\rightarrow sp^2$

Shape $\rightarrow$ v- shape

Uses :

(i) In refining petrol and sugar

(ii) In bleaching wool and silk

(iii) As an anti-chlor, disinfectant and preservative

(iv) In manufacturing of H_2SO_4, $NaHSO_3$

(iv) Liquid SO_2 is used as solvent.

Q.83 **SO_2 can acts as -**

(A) Reducing agent **(B) Oxidising agent**

(C) Bleaching agent **(D) Disinfectant**

Sol. **[A,B,C]**

$SO_2 + H_2O \rightarrow H_2SO_3$ Reducing nature

$SO_2 + H_2S \rightarrow 2H_2O + 3S$ oxidising nature

$SO_2 + H_2O \rightarrow H_2SO_4 + 2H$

colored matter + H $\rightarrow$ colorless matter

(Bleaching agent)

Q.84 **SO_3 on combining with HCl gives -**

(A) Chloro sulphonic acid **(B) Chlorine**

(C) SO_2Cl_2 **(D) None**

Sol. **[A]** $SO_3 + HCl \rightarrow SO_2.(OH)Cl$

Chloro sulphonic acid

Q.85 **Which one of the following on heating will give mixture of SO_2 and SO_3 ?**

(A) $ZnSO_3$ **(B) $CuSO_4$** **(C) Na_2SO_4** **(D) $FeSO_4$**

Sol. **[D]** $2FeSO_4 \xrightarrow[\text{Strong Heating}]{\Delta} SO_2 + SO_3 + Fe_2O_3$

Q.86 **When SO_2 is passed through acidified $K_2Cr_2O_7$ solution–**

(A) The solution turns blue **(B) The solution is decolourised**

(C) SO_2 is reduced **(D) Green $Cr_2(SO_4)_3$ is formed**

Sol. **[D]**

Acidified $K_2Cr_2O_7$ is turned green due to formation of chromic sulphate

$K_2Cr_2O_7 + 3SO_2 + H_2SO_4 \rightarrow K_2SO_4 + Cr_2(SO_4)_3 + H_2O$

green

Q.87 **Bleaching action of SO_2 is due to–**

(A) Reduction **(B) Oxidation**

(C) Hydrolysis **(D) Its acidic nature**

Sol. **[A]**

Bleaching action :- SO_2 in presence of moisture acts as a bleaching agent this is due to the reducing nature of SO_2

$SO_2 + 2H_2O \rightarrow H_2SO_4 + 2H$

Coloured Matter + H → Colourless matter

3. Hydrogen sulphide (H_2S) :

Preparation :

(i) By Treating FeS with sulphuric acid

$$FeS + H_2SO_4 \longrightarrow FeSO_4 + H_2S$$

This process occurs in kipp's apparatus

(ii) By HCl

$$Sb_2S_3 + 6HCl \longrightarrow 2SbCl_3 + 3H_2S$$

(Pure) (Pure)

By this method pure H_2S produced.

Properties :

(i) It is colouless gas with rotten egg smell.

(ii) It is moderately soluble in water but solubility decreases as increase temperature.

(iii) It act as reducing agent

Example :

(i) $H_2S + X_2 \longrightarrow 2HX + S$

(X = Cl, Br, I)

(ii) $H_2S + SO_2 \xrightarrow{\text{Moisture}} H_2O + S$

(iii) $2HNO_3 \longrightarrow H_2O + 2NO_2 + [O]$

$H_2S + [O] \longrightarrow H_2O + S$

$2HNO_3 + H_2S \longrightarrow 2H_2O + NO_2 + S$

Q.88 **H_2S on incomplete combustion with oxygen forms mainly –**

(A) H_2 and S **(B) H_2 and SO_2**

(C) H_2O and S **(D) H_2O and SO_2**

Sol. **[C]**

On incomplete combustion sulphur is formed.

$2H_2S + O_2 \rightarrow 2H_2O + 2S$

Q.89 **When H_2S is passed through nitric acid solution the products formed are–**

(A) b-sulphur (B) g-sulphur (C) colloidal sulphur (D) milk of sulphur

Sol. **[C]**

$2HNO_3 + H_2S \rightarrow 2NO_2 + S + 2H_2O$

Colloidal sulphur is formed.

Q.90 **H_2S cannot be dried by–**

(A) $CaCl_2$ **(B) P_2O_5** **(C) H_2SO_4** **(D) CaO**

Sol. **[C,D]**

$CaO + H_2S \rightarrow CaS + H_2O$

$H_2SO_4 + H_2S \rightarrow 2H_2O + SO_2 + S$

Q.91 **On passage of H_2S through an acidified $FeCl_3$ solution, $FeCl_3$ is converted to–** **[REE- 1991]**

(A) ferrous chloride **(B) ferric sulphide**

(C) ferrous sulphide **(D) ferrous sulphate**

Sol. **[A]**

If reduces ferric chloride to ferrous chloride

$2FeCl_3 + H_2S \rightarrow 2FeCl_2 + 2HCl + S$

Q.92 **Write the balanced chemical equations.** **[IIT - 1991]**
(a) Sulphur is precipitated in the reactions of hydrogen sulphide with sodium bisulphite solution.

Sol. $2H_2S + 2NaHSO_3 \rightarrow 3S^{-} + 3H_2O + Na_2SO_3$

Halogen Family

Elements	Electronic configuration	Nature
$_{9}F^{19}$ – Fluorine	[He] $2s^2\ 2p^5$	Non-metal
$_{17}Cl^{35}$ – Chlorine	[Ne] $3s^2\ 3p^5$	Non-metal
$_{35}Bl^{80}$ – Bromine	[Ar] $3d^{10}\ 4s^2\ 4p^5$	Non-metal
$_{53}I^{127}$ – Iodine	[Kr] $4d^{10}\ 5s^2\ 5p^5$	Non-metal
At–Astatine	[Xe] $4f^{14}\ 5d^{10}\ 6s^2\ 6p^5$	Non-metal

* At–Astatine is radioactive element.
Group-17 elements collectively known as halogens.
They are highly reactive non-metallic elements.

Atomic and Physical properties

1. Atomic Size (A.S.)

(a) Halogens have smallest atomic radius in their respective periods due to maximum effective nuclear charge

(b) As going down the group atomic size (Covalent radius) increases.

	F	<	Cl	<	Br	<	I
Covalent radius (pm)	64		99		114		133

2. Ionisation Energy (I.E.)

(a) Group-17 elements have very high ionisation energy.

(b) As going down the group ionisation energy decreases because atomic size increases.

	F	>	Cl	>	Br	>	I
I.E. (kJ / mole)	1680		1256		1142		1008

Electronegativity (E.N.)

(a) Electronegativity decrease as going down the group due to metallic character increases.

(b) Fluorine is the most electronegative element in periodic table.

E.N. ⇒	F >	Cl >	Br >	I
	4	3.2	3.0	2.7

Electron Gain Enthalpy [E.A.]

(a) Halogens have maximum negative electron gain enthalpy in their period.

(b) Fluorine has less electron gain Enthalpy compared to chlorine due to strong electronic repulsion in small 2p orbitals of fluorine.

	Cl >	F >	Br >	I
E.A. (kJ / mole)	−349	− 333	− 325	− 296

Physical State

F_2	Gas
Cl_2	Gas
Br_2	Corrosive liquid
I_2	Volatile Solid

Melting & Boiling Points

Their melting and boiling points increases with atomic number due to Vander waal forces increases down the group.

$F_2 < Cl_2 < Br_2 < I_2$

Vander waal force↑

b.p.	84.9	239.0	332.5	458.2
m.p.	54.4	172	265.8	386.6

7. Bond Energy (B.E.)

	F–F	Cl–Cl	Br–Br	I–I
Bond length (Å)	1.42	1.99	2.28	2.67
Bond Energy (kcal/mole)	38	57	45.5	35.6

Note : F_2 has small amount of B.E. due to greater inter electronic repulsion between lone pairs of small F-atom $:\ddot{F}—\ddot{F}:$

8. Colour

F_2 – light yellow
Cl_2 – Greenish yellow
Br_2 – Reddish brown
I_2 – Deep violet

Note: They show colour due to absorption of radiations in visible region which result in the excitation of outer electrons to higher energy levels.

9. Oxidation State

(a) All Halogens exhibit –1 oxidation state.
(b) Except F other halogens also show +1, +3, +5, +3 oxidation state due to vacant orbitals.
(c) They show +4 and +6 oxidation state in oxides and oxy acids.

10. Oxidising Power

Halogens act as strong oxidising agents due to greater reduction potential.

$$F_2 + 2e^- \rightarrow 2F^-; \quad E° = +2.87 \text{ volt}$$
$$Cl_2 + 2e^- \rightarrow 2Cl^- \quad E° = +1.36 \text{ volt}$$
$$Br_2 + 2e^- \rightarrow 2Br^-; \quad E° = +1.09 \text{ volt}$$
$$I_2 + 2e^- \rightarrow 2I^-; \quad E° = +0.54 \text{ volt}$$

SRP

Standard Reduction Potential

(a) F_2 is strongest oxidising agent even after the electron affinity of Fluorine is less than chlorine. This is because of its low bond dissociation energy and high heat of hydrogenate

$$\text{oxidising power} \Rightarrow \boxed{F_2 > Cl_2 > Br_2 > I_2}$$

11. Solubility

(a) Fluorine is highly soluble in water. it reacts with water to give mixture of O_2 and O_3.

$$2F_2 + 2H_2O \longrightarrow 4HF + O_2$$

$$3F_2 + 3H_2O \longrightarrow 6HF + O_3$$

(b) Other Halogens are soluble in organic solvents.

12. Hydration Energy of Halogen ions. (X^-)

Due to small size of ions hydration energy is higher.

F^-	Cl^-	Br^-	I^-	
515	381	347	305	(kJ/mole)

Q.93 The most reactive halogen is –

(A) I (B) Br (C) Cl (D) F

Sol. [D] F is most reactive due to least F– F Bond energy, highest electronegativity and highest reducing power of F atom.

Q.94 The halide which does not give a precipitate with $AgNO_3$ is –

(A) F^- (B) Cl^- (C) Br^- (D) I^-

Sol. [A] F^- does not give a precipitate with $AgNO_3$

Q.95 Which nonmetal does not combine directly with Cl_2, Br_2 and I_2–

(A) Carbon (B) Nitrogen (C) Oxygen (D) All of these

Sol. [D]

Because halogen and non metal both behave as same oxidising agent

Q.93 **Direction- The question consist of an 'assertion' and the 'reason'.**

(A) If both assertion and reason are correct, and reason is the correct ex planation of the assertion.

(B) If both assertion and reason are correct, but reason is not the correct explanation of the assertion.

(C) If assertion is correct but reason is incorrect.

(D) If assertion is incorrect but reason is correct.

(i) Assertion : Liquid I_2 conducts current very slightly
Reason : Iodine in liquid state show partial auto ionisation.

(ii) Assertion- F atom has a less negative electron affinity than Cl atom–
Reason- Additional electrons are repelled more effectively by 3p elec trons in Cl atom than by 2p electrons in F atom. **[IIT-1998]**

Sol. **(i) [C]**
Due to large size, it conducts electricity.
(ii) **[C]**

	F	Cl
electron affinity	– 3.6eV	– 3.8 eV

Assertion is correct but reason is false because additional electron in 2p subshell is more effectively repelled than 3p subshell in Cl atom.

Q.94 **Fluorine is the most reactive among all the halogens, because of its –** **[REE-1995]**

(A) Small size **(B) Low dissociation energy of F–F bond**
(C) Large size **(D) High dissociation energy of F–F bond**

Sol. **[A,B]**
Fluorine is most reactive due to
(i) Small size and high electronegativity
(ii) It is strongest oxidising agent
(iii) low dissociation energy of F–F bond.
$Cl_2 > Br_2 > F_2 > I_2$ (Bond dissociation energy)

Chemical Properties

1. Reactivity

$$\boxed{F_2 > Cl_2 > Br_2 > I_2}$$

Halogens are most reactive due to low bond dissociation energy, high electron affinity and high enthalpy of hydration.

2. Reactivity towards water

(a) F_2 decomposes water to O_2 and O_3

(b) Cl_2 and Br_2 decomposes water in the presence of sunlight.

$$2F_2 + 2H_2O \longrightarrow 4HF + O_2$$
$$3F_2 + 3H_2O \longrightarrow 6HF + O_3$$

$$Cl_2 + H_2O \longrightarrow HCl + \underset{\text{Hypochlorous Acid}}{HClO}$$
$$Br_2 + H_2O \longrightarrow HBr + \underset{\text{Hypobromous Acid}}{HBrO}$$

(c) I_2 does not decompose water.

3. Reaction with Hydrogen

By reaction with hydrogen halogen forms covalent hydrides.

$$H_2 + F_2 \longrightarrow 2HF\,(\text{very violent})$$
$$H_2 + Cl_2 \xrightarrow{\text{sunlight}} 2HCl$$
$$H_2 + Br_2 \xrightarrow[\text{pt–catalyst}]{\text{heat}} 2HBr$$
$$H_2 + I_2 \overset{\text{Heat}}{\rightleftharpoons} 2HI\,(\text{poor yield})$$

(i) Acidic Strength

In aqueous medium, HI is maximum acidic due to least Bond dissociation energy.

$$\boxed{HI > HBr > HCl > HF}$$

(ii) Reducing Nature

Due to less hydration energy, HI has High Reducing Nature

$$\boxed{HI > HBr > HCl > HF}$$

(iii) Boiling Point

$$\boxed{HF > HI > HBr > HCl}$$

Due to hydrogen bonding HF has maximum boiling point.

(iv) Thermal stability

$$\boxed{H-F > HCl > HBr > HI}$$

Due to small bond length of H–F, bond stability increases.

Q.95 **Which has the highest heat of vaporisation –**
(A) HF (B) HCl (C) HBr (D) HI

Sol. **[A]** HF has highest heat of vaporization due to strongest Hydrogen bonding

Q.96 **Which halogen acid is a liquid –**
(A) HF (B) HCl (C) HBr (D) HI

Sol. **[A]** HF is liquid

Q.97 **HBr/HI can be prepared by the reaction of –**
(A) Sodium, Br_2/I_2 and water **(B) Phosphorous, Br_2/I_2 and water**
(C) Potassium, Br_2/I_2 and water **(D) Selenium, Br_2/I_2 and water**

Sol. **[B]**
Bromine react with S, P, As.....
$2P + 3Br_2 \rightarrow 2PBr_3$
Iodine react with P, Fe, K
$2P + 3I_2 \rightarrow 2PI_3$

Q.98 **Boiling of dil. HCl acid does not increase its concentration beyond 22.2 percent because hydrochloric acid –**
(A) Is very volatile
(B) Is extremely soluble in water
(C) Forms a constant boiling mixture
(D) Forms a saturated solution at this concentration

Sol. **[C]**
Because it forms constant boiling mixture.

Q.99 **Anhydrous HCl is a bad conductor of electricity while aqueous HCl is a good conductor.**

Sol. In anhydrous state, HCl is covalent molecule. In aqueous solution, HCl combine with water molecules to form H_3O^+ and Cl^- ions.

$HCl + H_2O \rightarrow H_3O^+ + Cl^-$

Q.100 **Pure HI kept in a bottle acquires a brown colour after sometime.**

Sol. HI is strong reducing agent it is even oxidised by oxygen of the air the iodine is liberated which is dissolved imparting a brown colour to solution.

Q.101 **Ethyne is more soluble in HF because–**

(A) addition of HF to the multiple bond

(B) HF react with ethyne to yield CH_2F_2

(C) HF form H–bond with ethyne

(D) Ethyne is insoluble in HF

Sol. **[C]**

HF form H-Bond with ethyne.

4. Oxides and Oxyacids

(i) Oxides

Oxides are covalent in nature because low electronegativity difference between halogen and oxygen

$F \Rightarrow OF_2, O_2F_2$

$Cl \Rightarrow Cl_2O, ClO_2, Cl_2O_6, Cl_2O_7$

$Br \Rightarrow Br_2O, BrO_2, BrO_3$

$I \Rightarrow I_2O_5$

(ii) Oxiacids

Oxyacids	Oxidation state of Halogen	Oxyacids		
		Cl	Br	I
1. Hypohalous Acid	+1	HClO	HBrO	HIO
2. Halous Acid	+3	$HClO_2$	-	-
3. Halic Acid	+5	$HClO_3$	$HBrO_3$	HIO_3
4. Perhalic Acid	+7	$HClO_4$	$HBrO_4$	HIO_4

Note: Flourine form only one oxy Acid (HOF)

(a) All oxy Acid of halogens are monobasic

(b) Order of thermal stability of oxy Acids

$HClO_4 > HClO_3 > HClO_2 > HCl$

$HClO > HBrO > HIO$

$HClO_3 > HBrO_3 > HIO_3$

(a) Acidic character for same Halogen atom

$\overset{+7}{HClO_4} > \overset{+5}{HClO_3} > \overset{+3}{HClO_2} > \overset{+1}{HClO}$

(b) Acidic strength for different halogens.

$HClO > HBrO > HIO$

(c) Oxidising power of oxy Acids of chlorine is

$HClO_4 < HClO_3 < HClO_2 > HClO$

5. Reaction with Alkalies

$$2F_2 + \underset{\text{(cold dilute)}}{2NaOH} \longrightarrow 2NaF + OF_2 + H_2O$$

$$2F_2 + \underset{\text{(Hot conc.)}}{4NaOH} \longrightarrow 4NaF + O_2 + 2H_2O$$

(a) Halogens other than F_2 (means Cl_2, Br_2, I_2) react with NaOH as

$$X_2(g) + 2NaOH \longrightarrow NaX + NaOX + H_2O$$

$$X_2 + 2OH^- \xrightarrow{15°C} X^- + OX^- + H_2O$$

$$X_2(g) + 6OH^- \xrightarrow{70°C} 5X^- + XO_3^- + 3H_2O$$

6. Reaction with Halides

$$2KBr + Cl_2 \longrightarrow 2KCl + Br_2$$

$2KI + Cl_2 \longrightarrow 2KCl + I_2$

7. Bleaching action of halogens

$Cl_2 + H_2O \longrightarrow 2HCl + [O]$

Coloured matter + [O] $\longrightarrow$ Colourless

Q.102 Which of the following reaction is/are possible ?

(A) $K[BrICl] \xrightarrow{\Delta} KCl + IBr$ **(B)** $K[BrICl] \xrightarrow{\Delta} KBr + ICl$

(C) $K[BrICl] \xrightarrow{\Delta} KI + BrCl$ **(D) All of the above**

Sol. [A] $K[BrICl] \xrightarrow{\Delta} KCl + IBr$

Q.103 Pick out the incorrect statement regarding halogens –

(A) Aq. solutions of Cl_2, Br_2 or I_2 oxidize H_2S to S

(B) Chlorine displaces bromine and iodine from aqueous solution of bro mides and iodides

(C) Cl_2, Br_2 and I_2 all oxidize sodium thiosulphate to sodium sulphate

(D) Iodine is converted into silver iodide which is used in the manufac ture of photographic films.

Sol. [C]

Cl_2, Br_2 or I_2 is more oxidising agent than H_2S, thus it can accept easily electron from H_2S and oxidize H_2S to S.

$\Rightarrow$ I^-
Cl^-
Br^-
OH^-
NO_3^-
SO_4^{2-}
PO_4^{3-}
CO_3^{2-}
F^-

According to series lower element displaces to Upper element because lower element is more oxidising agent than Upper element thus it can accept electron from upper elements and oxidize to upper elements.

$\Rightarrow$ $Na_2S_2O_3 + H_2O + Cl_2 \rightarrow Na_2SO_4 + S + 2HCl$
$Na_2S_2O_3 + H_2O + Br_2 \rightarrow Na_2SO_4 + S + 2HBr$

$2Na_2S_2O_3 + I_2 \rightarrow 2NaI + Na_2S_4O_6$
sodium
tetrathionate

AgI is used in photographic film.

Q.104 **Fluorination of organic compound is possible with fluorine –**

(A) Concentrated with nitrogen in the presence of a reducing agent

(B) Diluted with nitrogen in the presence of a reducing agent

(C) Diluted with nitrogen in the presence of a catalyst

(D) Diluted with oxygen in the presence of a catalyst

Sol. **[C]**

N_2 have more bond dissociation energy (225 kcal/mole) than O_2 (118.3 kcal/mole) Thus in case of N_2 more amount of heat is release . Thus in case of diluted this reaction is possible.

Q.105 **KF combines with HF to form KHF_2. The compound contains the species –**

(A) K^+, H^+, and F^- **(B) K^+, F^- and HF^-**

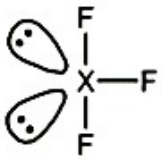
(C) K^+ and $[HF_2]^-$ **(D) $[KHF]^+$, and F^-**

Sol. **[C]**

$KF + HF \rightarrow KHF_2 \rightarrow K^+ + [HF_2]^-$

8. Inter Halogen Compounds

The main reason for the formation of these compounds is the electronegativity and size difference among the Halogens.

	AX	AX_3	AX_5	AX_7
Example	ClF, BrF, IF, BrCl, ICl, IBr	ClF_3, BrF_3, IF_3, ICl_3	ClF_5, BrF_5, IF_5	IF_7
Geometry	Linear	 T-shaped	 Square Pyramidal	 Pentagonal bipyramidal
Hybridisation	sp^3	sp^3d	sp^3d^2	sp^3d^3

Note: Fluorine has the most electronegativity and smallest size, so it forms maximum, number of inter halogens compounds.

Properties

(a) They are covalent compounds due to little electronegativity difference between different halogens.

(b) Reactivity is high due to weakness of covalent bond.

(c) Their stability increases as the size of centre atom increases.

(d) They are diamagnetic because their valence electrons are and present either as shared or unshared electron pair.

Q.106 **Inter halogen compounds are more reactive than the individual halogen because –**

(A) Two halogens are present in place of one

(B) They are more ionic

(C) Their bond energy is less than the bond energy of the halogen molecule

(D) They carry more energy

Sol. **[C]**

Interhalogen compound (AB) A always bigger atom and B smaller atom (ClF, ClF_3, IF_5, IF_7).

AB type compounds are more reactive than A_2 or B_2 molecule since A–B Bond is weaker than A–A and B–B bonds.

A—B	A—A or B—B
→ Low bond energy	→ more bond energy
→ more reactive bond	→ less reactive

Q.107 **Which of the following is/are not characterise inter halogen compounds ?**

(A) Inter halogen compounds are reactive than ordinary diatomic halogens.

(B) The order of reactivity is $ClF_3 > BrF_5 > IF_7$

(C) $BrF_5 \xrightarrow{H_2O} HBrO_3 + HF$

(D) All of the above

Sol. **[D]** Being polar, in halogen compound are more reactive than halogen (except F_2). The general order of reactivity.

$ClF_3 > BrF_5 > IF_7 > BrF_3 > IF_5 > BrF$

$BrF_5 \xrightarrow{H_2O} HBrO_3 + H^+$

9. Poly Halides

Halogen or interhalogen halides combine to give poly halides.

Example:

(a) In presence of KI, I_2 is more soluble in water so form KI_3. It is due to formation I_3^-.

$I^- + I_2 \longrightarrow I_3^-$

$KI + I_2 \longrightarrow kI_3$

(b) $Cl_3^-, Br_3^-, ICl_2^-, IBr_2^-, I_3^-$
Geometry = linear
Hybridisation = sp^3d

(c) $Cl^{3+}, Br_3^+, I_3^+, ICl_2^+, IBr_2^+$
Geometry = Bent Shape
Hybridisation = sp^3

(d) ICl_4^-, BrF_4^-, I_5^-
Hybridisation – sp^3d^2
Geometry – Square planar

(e) I_7^+
Hybridisation = sp^3d^2
Geometry = octahedral

Q.108 Fluorine does not form any polyhalide as other halogens because –

(A) It has maximum ionic character

(B) It has low F–F bond energy (38.5 kcal mol^{-1})

(C) Of the absence of d-orbitals in the valence shell of fluorine

(D) It brings about maximum coordination number in other elements

Sol. [C]

Fluorine does not have vacant "d" orbital in valence shell therefore it doesn't combine with F^- ion to form poly fluoride ions like Cl_3^-, Br_3^-, I_3^- etc.

10. Formation of iodine Cations

Due to lesser ionisation energy, iodine forms compounds with I^+ and I^{+3} cations

Example:

IPO_4

Here $IPO_4 \rightleftharpoons I^{3+} + PO_4^{-3}$ (electrolysis)

At Cathode we get I_2.

11. Pseudo halogens and Pseudohalides

A pseudo halogen is made up of two E.N. atoms/group and in its molecular form resembles the halogens.

(a) Pseudo halogens behave like halogen in chemistry

Pseudo Halogens	**Formula**
Cyanogen	$(CN)_2$
Oxo Cyanogen	$(OCN)_2$
Thio Cyanogen	$(SCN)_2$
Seleno Cyanogen	$(SeCN)_2$

Pseudo Halides	Formula
Cyanide	CN^-
Cyanate	OCN^-
Thiocyanate	SCN^-
Selenocyanate	$SeCN^-$

Q.109 **CN^- is pseudo halides, when cyanogens are heated with alkali, the products are–**

(A) HCN, H_2O **(B) NH_3, $(NH_4)_2CO_3$**

(C) NaCN, NaOCN **(D) HCOONa, NH_3**

Sol. **[C]**

$Na + (CN)_2 \xrightarrow{H_2O} NaCN + NaOCN$

Q.110 **The structure of azido carbondisulphide is –**

(A) $CS_2(N_3)_2$ **(B) $(SCSN_3)_2$** **(C) CS_2N_3** **(D) $(CSN_3)_2$**

Sol. **[B]** Structure of azido carbondisulphide is $(S\ CSN_3)_2$

12. Freons (CFCs)

Some of Freons are cause of ozone depletion.

Freon –11 is CCl_3F

Freon –12 is CCl_2F_2

Freon –113 is $CCl_2F \cdot CClF_2$

Freon –115 is $CClF_2 \cdot CF_3$

Anomalous Behaviour of Fluorine

Fluorine show different behaviour from rest of element due to

(a) Its small size

(b) Its high Electronegativity

(c) Absence of d-orbitals

(d) Low dissociation energy of F_2 molecule

(e) Some of properties which differ from rest of the halogens

(i) Reactivity $\Rightarrow$ very high

(ii) Oxidation state $\Rightarrow$ –1 only

(iii) Hydrogen bonding present in fluorine molecules

(iv) Weak Acidic character
(v) Nature of compound is ionic

(vi) It does not form poly halides like F_3^-
(vii) Solubility: AgF is soluble in water while AgCl, AgBr and AgI are not soluble due to high covalent nature of AgX while AgF is ionic in Nature

Halogens preparation and Uses

1. Fluorine (F_2)

(a) Fluorine is found only in combined state
(b) Main ore of fluorine are
Fluorospar – CaF_2
Cryolite – Na_3AlF_6
Opatite – $\left[CaF_2 \cdot 3Ca_3(PO_4)_2\right]$
(c) Isolation of F_2 cannot be carried out easily due to following reasons.
(i) F_2 do not react with glass, platinum, carbon.
(ii) F_2 is a good oxidising agent so generally oxidising agents are not able to oxidise F^- to F_2.

Preparation

(a) Fluorine prepared in special container which is made by Monel (Alloy of Ni–Cu–Fe).
(b) If is obtained by electrolysis of molten KHF_2 including copper. However a thin layer of metal fluoride formed on the inner metallic surface resist further attack and protects the vessel from corrosion.

$$2KHF_2 \xrightarrow[\text{(molten)}]{} 2KF + 2HF$$

or

$$2KHF_2 \longrightarrow 2K^+ + 2H^+ + 4F^-$$

At Cathode

$$2H^+ + 2e^- \longrightarrow H_2$$

At Anode

$$4F^- \longrightarrow 2F_2 + 4e^-$$

Properties

(a) It is pale yellow gas
(b) It is strong oxidising agent

Uses

(a) UF_6 in nuclear power plants
(b) SF_6 is used as an insulator
(c) Used in CFC and Teflon
(Chloro fluoro Carbon)

Q.111 **Standard electrode potential is highest for –**

(A) $\frac{1}{2}F_2(g) + e^- \rightleftharpoons F^-(aq)$ **(B)** $\frac{1}{2}Cl_2(g) + e^- \rightleftharpoons Cl^-(aq)$

(C) $\frac{1}{2}Br_2(g) + e^- \rightleftharpoons Br^-(aq)$ **(D)** $\frac{1}{2}I_2(g) + e^- \rightleftharpoons I^-(aq)$

Sol. **[A]**

$$\frac{1}{2}F_2(g) + e^- \rightleftharpoons F^-(aq)$$

zero → Reduction → (–1) oxidation state

Due to small size and high electronegativity its reduction potential have maximum value.

Q.112 **Fluorine cannot be prepared from fluorides by chemical oxidation. Why?.**

Sol. The standard reduction potential of fluorine is maximum.

$$:\ddot{\underset{\cdot\cdot}{F}}\cdot + e^- \longrightarrow :\ddot{\underset{\cdot\cdot}{F}}:^-$$

Thus, it can't be oxidised by any other reagent. F^- ion is very stable due to small size and high electronegativity of fluorine atom.

Q.113 **KF combines with HF to form KHF_2. The compound contains the species –**

(A) K^+, H^+, **and** F^- **(B)** K^+, F^- **and** HF^-

(C) K^+ **and** $[HF_2]^-$ **(D)** $[KHF]^+$, **and** F^-

Sol. **[C]**

$KF + HF \rightarrow KHF_2 \rightarrow K^+ + [HF_2]^-$

2. **Chlorine (Cl_2)**

It is found in combined state and sea water is its main source.

Ore ⇒ Carnelite $KCl \cdot MgCl_2 \cdot 6H_2O$

Horn silver AgCl

Preparation

(a) Reaction of HCl with oxidising agent like $KMnO_4, MnO_2, K_2Cr_2O_7, PbO_2, CaOCl_2$ and O_3 etc.

$$2KMnO_4 + 16HCl \longrightarrow 2KCl + 2MnCl_2 + 8H_2O + 5Cl_2$$

$$MnO_2 + 4HCl \longrightarrow MnCl_2 + 2H_2O + Cl_2$$

$$Ca(OCl)Cl + H_2SO_4 \longrightarrow CaSO_4 + H_2O + Cl_2$$

(b) By electrolysis of brine solution

$$NaCl \longrightarrow Na^+ + Cl^-$$

At Anode

$$Cl^- \longrightarrow \frac{1}{2}Cl_{2(g)} + 1e^-$$

At Cathode

$$Na^+ + 1e^- \longrightarrow Na_{(s)}$$

Properties

(a) It is greenish yellow gas with a suffocation smell, poisonous in nature.
(b) It is 2.5 times heavy as air
(c) It is soluble in water, give yellowish strongly smelling liquid called chlorine
(d) It has oxidising and bleaching properties

$$Cl_2 + H_2O \longrightarrow HCl + HOCl$$

$$HOCl \longrightarrow HCl + [O]$$

Colour matter + [O] $\longrightarrow$ colourless matter

(e) $2KI + Cl_2 \longrightarrow 2KI + I_2$

It reacts with different Halides

(f) Action of Alkali on Cl_2

$$\underset{\text{(cold)}}{2NaOH} + Cl_2 \longrightarrow NaCl + \underset{\text{Sodium hypochlorite}}{NaOCl} + H_2O$$

$$6NaOH + 3Cl_2 \longrightarrow 5NaCl + \underset{\text{Sodium chlorate}}{NaClO_3} + 3H_2O$$

(g) Reaction with $Ca(OH)_2$

Chlorine reacts with Calcium hydroxide to give bleaching powder

$$Ca(OH)_2 + Cl_2 \longrightarrow CaOCl_2 + H_2O$$

(h) Reaction with Ammonia

$$\underset{\text{(excess)}}{8NH_3} + 3Cl_2 \longrightarrow 6NH_4Cl + N_2$$

$$NH_3 + \underset{\text{(excess)}}{3Cl_2} \longrightarrow NCl_3 + 3HCl$$

(i) Formation of Addition Compounds

$$SO_2(dry) + Cl_2 \longrightarrow \underset{\text{Sulphuryl Chloride}}{SO_2Cl_2}$$

$$CO + Cl_2 \longrightarrow COCl_2\ (\text{Phosgene})$$

or

Carbonyl Chloride

(j) Action on organic compounds

$$CH_4 + Cl_2 \longrightarrow \underset{\text{Substitution Reaction}}{CH_3Cl} + HCl$$

Uses

(a) Bleaching Agent
(b) Used in germicide and disinfectant
(c) In manufacturing of Antiseptic like NaOCl
(d) In extraction of gold and Pt.
(e) In manufacturing of HCl, bleaching powder CCl_4, CH_2Cl_2, DDT, rubbers etc.

Q.114 **$Cl_2(g)$ can be prepared by heating of –**

(A) NaCl **(B) $NaClO_4$** **(C) $PtCl_4$** **(D) $PbCl_4$**

Sol. **[C,D]**

$PtCl_4 \xrightarrow{\Delta} PtCl_2 + Cl_2\uparrow$ $PbCl_4 \xrightarrow{\Delta} PbCl_2 + Cl_2$ gas$\uparrow$

Q.115 **(X) $CaOCl_2$ Produces Cl_2 in moist air ?**

(A) due to the reaction with water
(B) due to the reaction with CO_2
(C) both (A) & (B) are correct
(D) because of disproportionation of (X)

Sol. **[C]**

$CaOCl_2 + CO_2 \rightarrow CaCO_3 + Cl_2$
$CaOCl_2 + H_2O \rightarrow Ca(OH)_2 + Cl_2$

3. Bromine (Br_2)

It is found only in combined state

Source $\Rightarrow$ Seawater and Carnelite contain 0.01 to 1% $MgBr_2$.

Preparation

(a) Lab method

$$2KBr + 3H_2SO_4 + MnO_2 \longrightarrow 2KHSO_4 + MnSO_4 + 2H_2O + Br_2$$

(b) $2KBr + Cl_2 \longrightarrow 2KCl + Br_2$

(c) From carnallite – it's aqueous solution to given $MgBr_2$ with Cl_2

$$MgBr_2 + Cl_2 \longrightarrow MgCl_2 + Br_2$$

Properties

(a) It is reddish brown heavy liquid and it is poisonous

(b) Soluble in water

$$H_2O + Br_2 \longrightarrow HOBr \longrightarrow HBr + [O]$$

(c) Oxidising and bleaching action same as Cl_2

Uses

(a) Use as oxidising agent

(b) In preparation of ethyl bromide (C_2H_5Br)

(C_2H_5Br is used in manufacturing of TEL)

TEL ⇒ Tetraethyl lead an important Anti knoking compound in petroleum industry

$$C_2H_6 + Br_2 \longrightarrow C_2H_5Br + HBr$$

$$4C_2H_5Br + 4NaPb \longrightarrow (C_2H_5)_4 Pb + 4NaBr + 3Pb$$

Q.116 **Bromine vapour turns moist starch-iodide paper–**

(A) Brown (B) red (C) Blue (D) Colourless

Sol. **[C]**

Bromine vapour turns moist starch iodide paper blue.

Q.117 **Identify (A) and (B).**

$$\mathbf{Br_2 + OH^- \rightarrow (A) + (B)}$$

hot

$$\mathbf{(A) + (B) + H^+ \rightarrow Br_2}$$

(A) gives yellow ppt. with $AgNO_3$

Sol. $3Br_2 + 6OH^- \rightarrow \underset{(A)}{5Br^-} + \underset{(B)}{BrO_3^-}$

$5Br^- + BrO_3^- + 6H^+ \rightarrow 3Br_2 + 3H_2O$

(A) $Br^- + Na \rightarrow NaBr$

$NaBr + AgNO_3 \rightarrow AgBr + NaNO_3$

(A) yellow ppt.

Q.118 **A liquid A is treated with Na_2CO_3 solution. A mixture of two salts B and C are produced in the solution. The mixture on acidification with sulphuric acid and distillation produces the liquid A again. Identify A, B and C and write the equations involved –** **[IIT-1997]**

Sol. $\underset{(A)}{3Br_2} + 3Na_2CO_3 \rightarrow \underset{(B)}{5NaBr} + \underset{(C)}{NaBrO_3} + 3CO_2$

$5NaBr + NaBrO_3 + 3H_2SO_4 \rightarrow 3Na_2SO_4 + \underset{(A)}{3Br_2} + 3H_2O$

4. **Iodine (I_2)**

It is rare, occur in combined state

Source ⇒ Unwated sea plant kelp, ash obtained on burning sea-weeds (1% iodine)

Preparation

(a) Lab method

$2KI + H_2SO_4 \xrightarrow{MnO_2} K_2SO_4 + SO_2 + I_2 + 2H_2O$

(b) Commercially from sea weeds

Properties

(a) It is a black (dark violet) shining solid.

(b) It is least soluble in water but soluble in solution of KI due to formation of I_3^-

$KI + I_2 \longrightarrow KI_3$

(c) 2-3% solution of I_2 with C_2H_5OH act as antiseptic and called Tincture of iodine.

(d) Soluble in organic solvents like CS_2

(e) Action with Alkali on I_2 is same like Cl_2 and Br_2

(f) Action with Ammonia

$2NH_3 + 3I_2 \longrightarrow NI_3 \cdot NH_3 + 3HI$ explosive

$8NI_3.NH_3 \longrightarrow 5N_2 + 9I_2 + 6NH_4I$

(g) Reaction with sodium thiosulphate

$2Na_2S_2O_3 + I_2 \longrightarrow Na_2S_4O_6 + 2NaI$

While Cl_2 gives $Na_2S_2O_3$ and Br_2 gives S.

Uses

(a) Largely used in medicine in iodex, iodoform

(b) Used as laboratory reagent

(c) Iodine does not act as a bleaching agent but used as disinfectant

Q.119 **When I_2 is dissolved in aqueous solution of KI then colour changes from brown to deep red. What molecular species exist in the solution and what is its structure ? Write down more compounds having similar geometries.**

Sol. $KI + I_2 \rightarrow KI_3$

Brown Deep Red

Colour

$KI_3 \rightarrow K^+ + I_3^-$

I_3^- species exist in solution

$[I - I - I]^- I_3^-$ has linear shape other molecules having similar geometry

e.g. XeF_2, IBr_2^- etc.

Q.120 **The brown colour of an acidified dilute solution of iodine in aqueous KI is intensified by the addition of nitrite but it is discharged by the addition of a sulphite.**

Sol. It is slightly soluble in water giving a light brown solution. Its solubility increases in water in presence of potassium iodide due to formation of potassium tri-iodide. KI_3 easily breaks down and thus the solution has all the properties of free iodine.

$KI + I_2 \rightleftharpoons KI_3$

Q.121 **Chlorine is gas while iodine is solid under ordinary conditions.**

Sol. Chlorine is gas while iodine is solid because in halogen with moleculer mass increases Vander wall's force increases up to down moleculer mass increases thus, Vander wall's force also increases.

Thus F_2 Cl_2 Br_2 I_2

gases volatile liquid volatile solid

Comparison of [Cl_2, Br_2 and I_2]

S. No.	Property	Chlorine	Bromine	Iodine
1.	Physical State	Gas	Liquid	Solid
2.	Colour of Vapour	Greenish Yellow	Dark Red	Violet
3.	Action of H_2O	Decomposes into HCl & O_2	Decomposes Slowly in Presence of Light	No action
4.	Oxidising Action	Strong	Good	Weak
5.	Bleaching Action	Moist Cl_2 is a Good Bleaching Agent	Moist Br_2 is a good Bleaching Agent	No Bleaching
6.	Action of Halides	Displaces Br_2 & I_2	Displaces I_2	No Action
7.	Combination with H_2	Explosive in Light Slow in Dark	Only on Heating	Heating + Catalyst

5. Hydrogen Halides

Preparation

(a) $H_2 + F_2 \xrightarrow{dark} 2HF$

(b) $H_2 + Cl_2 \xrightarrow{hv} 2HCl$

(c) $H_2 + Br_2 \xrightarrow[473K]{Pt} 2HBr$

(d) $H_2 + I_2 \xrightarrow[710k]{Pt} 2HI$

Commercially Halides are formed as

(a) HF is formed by

$CaF_2 + \text{Conc. } H_2SO_4 \xrightarrow{\Delta} CaSO_4 + 2HF_{(g)}$

$\Rightarrow HF_{(g)}$ is dissolve in water to given HF

(b) HCl is formed by

$2NaCl + \text{Conc. } H_2SO_4 \xrightarrow{\Delta} Na_2SO_4 + 2HCl_{(g)}$

(c) HBr and HI are strong reducing agent so they are not formed by concentration H_2SO_4 we use phosphoric Acid

$3NaBr + H_3PO_4 \xrightarrow{\Delta} Na_3PO_4 + 3HBr$

Properties

(a) Acidic strength of Halides

HI > HBr > HCl > HF

(b) At normal temperature HI is liquid while other are gas

(c) Fluorine $\Rightarrow KHF_2$ type salt exist due to F has high electro negativity and small size so here H-Bonding takes place.

$HF_2^{\ominus}$ ion $\Rightarrow (F - H - - - F^{\ominus})$

Q.122 HCl cannot form H_2Cl_2, while HF can form H_2F_2. The reason is –

(A) Fluorine is more reactive

(B) HF is more reactive

(C) Fluorine atom is small and can form hydrogen bonds **(D) None**

Sol. [C]

More electronegative atom such as N, O and F form hydrogen bonding thus. HCl can't form but HF can form hydrogen bonding.

Q.123 The mixture of concentrate HCl and potassium chlorate on heating gives – [IIT-2000]

(A) Cl_2 only **(B) ClO_2 only**

(C) $Cl_2 + ClO_2$ **(D) $Cl_2 + ClO_2 + ClO_3$**

Sol. [C]

$2KClO_3 + 4HCl \rightarrow 2KCl + 2ClO_2 + Cl_2 + 2H_2O$

The mixture of ClO_2 and Cl_2 is called euchlorine.

Q.124 $K_2Cr_2O_7 + HCl \rightarrow KCl + CrCl_3 + ...+ .....$ **[IIT-1992]**

Sol. $K_2Cr_2O_7 + 14\ HCl \rightarrow 2KCl + 2\ CrCl_3 + 7H_2O + 3Cl_2$

6. Bleaching Powder ($CaOCl_2$)

Preparation

$$Ca(OH)_{2(s)} + Cl_{2(g)} \longrightarrow CaOCl_{2(s)} + H_2O_{(g)}$$

Properties

(a) It is yellow powder with smell of Cl_2

(b) It loses its Cl_2 by the action of dilute Acid or CO_2

$$CaOCl_2 + 2HCl \longrightarrow CaCl_2 + H_2O + Cl_2$$

$$CaOCl_2 + CO_2 \longrightarrow CaCO_3 + Cl_2$$

(c) It loses its oxygen in presence of Cobalt Chlorine ($CaCl_2$), which act as catalyst

$$2CaOCl_2 \xrightarrow{CoCl_2} 2CaCl_2 + O_2$$

Uses

(a) It is used for bleaching pulp, cotton linen etc.

(b) For making wood unshrinkable

(c) For making chloroform

Paragraph for Q.No. 128 to 130

A sample of (X) was kept for one year in an open container; then old sample of (X) was used after drying. The dried sample of (X) is divides into two equal parts. To the first part excess of KI and HCl has been added which liberates I2. The liberated I2 require 20 ml of 0.1(N) Na2S2O3. To the other part excess of KI and of 0.01(N) Na2S2O3. To the other part excess of KI and CH3COOH has been added the liberated I2 require 19 ml of 0.1(N) Na2S2O3.

Q.125 **As per the above observation , (X) is–**

(A) Ca(OCl)Cl **(B) $CaCl_2$**

(C) $Ca(ClO_3)_2$ **(D) $Ca(OH)_2$**

Sol. **[A]**

Inorganic compound (X) is $CaOCl_2$

Q.126 **A sample of (X) was kept for one year in an open container; then old sample of (X) was used after drying. The dried sample of (X) is divides into two equal parts. To the first part excess of KI and HCl has been the liberated I_2 require 20 ml of 0.1(N) $Na_2S_2O_3$. To the other part excess of KI and of 0.01(N) $Na_2S_2O_3$. To the other part excess of KI and CH_3COOH has been added the liberated I_2 require 19 ml of 0.1(N) $Na_2S_2O_3$.**

Explain why old sample of (X) react different in the presence of CH_3COOH & HCl ?

(A) old sample of (X) lost some amount of chlorine

(B) old sample of (X) has been hydrolysed to some extent

(C) old sample of (X) has been partially converted to $CaCl_2$

(D) old sample of (X) has been partially converted to $CaCl_2$ & $Ca(ClO_3)_2$.

Sol. **[D]**

$CaOCl_2$ divided into two part → (i) $CaOCl_2$, (ii) $CaOCl_2$

(i) $CaOCl_2 + 2CH_3COOH + 2KI \rightarrow Ca(CH_3COO)_2 + 2KCl + H_2O + I_2$

(ii) $CaOCl_2 + 2KI + 2HCl \rightarrow CaCl_2 + H_2O + 2NaCl + I_2$

$6CaOCl_2 \xrightarrow{\text{Autooxidation}} 5CaCl_2 + Ca(ClO_3)_2$

Q.127 **Sample (X) act as an moderate oxidising agent because–**

(A) it gives nascent oxygen **(B) it gives nascent chlorine**

(C) it gives $HClO_3$ **(D) it produce chlorine gas on hydrolysis**

Sol. **[A]**

$CaOCl_2 \rightarrow CaCl_2 + O$

nascent oxygen

Noble gases [Inert Gas]

Elements	Electronic configuration	Nature
He-Helium	$1s^2$	
Ne-Neon	$[He]2s^22p^6$	
Ar-Argon	$[Ne]3s^23p^6$	
		Gaseous
Kr-Krypton	$[Ar]3d^{10}4s^24p^6$	
Xe-Xenon	$[Kr]4d^{10}5s^25p^6$	
Rn-Radon	$[Xe]4f^{14}5d^{10}6s^26p^6$	

- Group-18 elements (Zeroth group) are called as inert/Nobel gas. Because they are chemically inert and found in gaseous state.
- Xe and Rn are rare and Radon is Radioactive and obtained by decay of Radium (Ra^{226}).

$$^{226}Ra_{88} \longrightarrow {}^{222}_{86}Rn + {}^{4}_{2}He$$

- Most Abundance in Air is Ar.
  Abundance of Nobel gas elements.

  Ar > Ne > Kr > He > Xe

Physical Properties

1. **Atomic Radius : -** As going down the group atomic radius is increases.
   Note : - Generally for Nobel gases we observe Vander wall radius .

   He < Ne < Ar < Kr < Xe

   ⟶ Atomic Radius ↑
2. **Ionisation energy : -** Nobel gas are chemically inert and do not have tendency to loss are gain of electron due to complete octet. Therefore, they have maximum

ionisation energies in periodic table (along there are corresponding period).

- In group ionisation energy decreases going down the group.

$$\boxed{He > Ne > Ar > Kr > Xe}$$
$$\longrightarrow \text{Ionisation Energy} \downarrow$$

3. **Electron Affinity (E.A.) : -** Noble gases have more positive electron gain Affinity due to complete octet.

4. **Physical state : -** Except radon all elements are gaseous in Atmosphere and they are colourless and Tasteless.

5. **Melting and boiling point : -** Nobles gas have very low meting and boiling point because they have weak Vander wall forces.

   - Helium has the lowest boiling point of any know substance.

6. **Solubility : -** Noble gases are sparingly soluble in water.

Physical properties :

Elements	Atomic radius (PM)	Ionisation Energy (Kj/mol)	Boiling Point (K)
$_{2}He^{4}$	120	2372	4.2
$_{10}Ne^{20}$	160	2080	27.1
$_{18}Ar^{40}$	190	1520	87.2
$_{36}Kr^{81}$	200	1351	119.7
$_{54}Xe^{131}$	220	1170	165.0

Q.128 Helium is obtained from which of the following –

(A) Natural gases trapped under rock formations (B) Liquid air

(C) Radioactive decay (D) Gasoline

Sol. (A,C)

Q.129 The inert gas abundantly found in atmosphere is –

(A) Xe (B) Kr (C) Ar (D) He

Sol. (C)

Q.130 The word argon means –

(A) Noble (B) Now (C) Strange (D) Lazy

Sol. (D)

Q.131 Which of the following is most polarised among Noble gases –

(A) He (B) Xe (C) Kr (D) Rn

Sol. (D)

Q.132 Electron affinity for a noble gas is approximately equal to –

(A) That of halogens (B) Zero

(C) That of oxygen family (D) That of nitrogen family

Sol. (B)

Q.133 **Helium has which of the following properties –**

(A) Has the lowest boiling point of any substance
(B) Does not freeze to a solid at any temperature
(C) Able to flow without viscosity below 2K
(D) Able to exist in more than one liquid phases

Sol. **(A,B,D)**

Q.133 **The noble gas which shows abnormal behaviour in liquid state and behave as superfluid is –**

(A) Ne **(B) He** **(C) Ar** **(D) Xe**

Sol. **(B)**

Q.134 **Noble gases are paramagnetic in nature.**

Sol. False.
All electron are paired thus it is diamagnetic.

Q.135 **Helium cannot be adsorbed by charcoal.**

Sol. True.
Due to high ionisation energy.

Q.136 **All the noble gases are present in the atmosphere in very minute quantities.**

Sol. True.
It is present in the atmosphere to the extent of 1 % by volume.

Q.137 **Which noble gas is the least polarisable–**

(A) He **(B) Ne** **(C) Kr** **(D) Rn**

Sol. **(A)** Polarisability $\propto$ size

$$\underset{\text{Polarisability increases}}{\text{He Ne Ar Kr Xe} \longrightarrow}$$

Q.138 **The solubility of noble gases in water shows the order –**

(A) He > Ar > Kr > Ne > Xe **(B) He > Ne > Ar > Kr > Xe**

(C) Xe > Kr > Ar > Ne > He **(D) None**

Sol. **(C)** Solubility µ size of noble gases µ polarisability
He < Ne < Ar < Kr < Xe (solubility)

Q.139 **The forces of cohesion in liquid helium are–**

(A) Covalent **(B) Ionic** **(C) Vander waal's** **(D) Metallic**

Sol. **(C)** Between all inert gases (noble gases) Wander wall' s force exists.

Q.140 **Noble gases are –**

(A) Colourless **(B) Odourless**

(C) Tasteless and non inflammable **(D) All**

Sol. **(D)** Noble gases –
(i) colourless
(ii) odourless
(iii) tasteless and non inflammable

Chemical Properties

Noble gases are least reactive (inert) due to –

- Their octet is compete (ns^2np^6) except helium ($1s^2 \rightarrow$ form duplet).
- High ionisation Energy and most positive electron affinity.

Note : It is observed that xenon first ionisation energy (1170 kJ/mole) is almost equal to first ionisation energy of oxygen therefore xenon form same compound, mainly with high electronegative element like fluorine and oxygen few compounds of krypton and radon are also found. Burt in case of He, Ne, Ar no true compounds are yet known.

Compounds of Xenon

As we discussed above that Xenon form compound with high electronegative elements like fluorine and oxygen.

A. **compounds of Xenon with fluorine :**

1. **Xenon difluoride (XeF_2)**

Preparation :

(i) By direct reaction with fluorine

$$\underset{\text{(excess)}}{Xe_{(g)}} + F_{2(g)} \xrightarrow[\text{monel metal}]{873k,\ 1bar} XeF_{2(s)}$$

(ii) By reaction with O_2F_2

$$Xe_{(g)} + O_2F_2 \xrightarrow{118^\circ C} XeF_2 + O_2$$

This reaction is due to O_2F_2 is not stable and easily react with Xe and for XeF_2.

(iii) By Electric arc method :

$$Xe + F_2 \xrightarrow{Hg(arc)} XeF_2$$

(iv) Recently discovered method :

In this method first we produce $K[AgF_4]$ which further react with BF_3 and give AgF_3. On treating Xe with AgF_3, Xenon fluoride will form.

$$K\left[AgF_4\right] \xrightarrow{BF_3} \underset{\text{(red solid)}}{AgF_3}\downarrow + KBF_4$$

$$2AgF_3 + Xe \longrightarrow \underset{\text{(Brown solid)}}{2AgF_2}\downarrow + XeF_2$$

Properties :

(1) It is colourless crystallin solid

(2) It is soluble in HF liquid and dissolve in water to give solution with a pungent odour.

(3) It is stored in container of Monel metal (alloy of nickel)
(4) Reaction with hydrogen (H_2) :

$$XeF_2 + H_2 \longrightarrow Xe + 2HF$$

(5) Hydrolysis of XeF_2 :
(a) Direct hydrolysis is also due to formation HF because XeF_2 dissolve in HF easily and reversible reaction.

$$2XeF_2 + 2H_2O \overset{\text{Slow}}{\rightleftharpoons} 2Xe + HF + O_2$$

(b) In presence of Alkali Hydrolysis is fast.

$$XeF_2 + 2NaOH \xrightarrow{\text{Fast}} Xe + 1/2\ O_2 + 2NaF + H_2O$$

(6) Oxidising Nature :
Due to more positive value of standard reduction potential of XeF_2 it acts as a strong oxidising agent.

$$XeF_2 + 2H^+ + 2e^- \longrightarrow Xe + 2HF; E^\circ = 2.64$$

Example :
(i) Oxidises halides (except fluorides)

$$XeF_2 + 2HCl \longrightarrow Xe + 2HF + Cl_2$$
$$XeF_2 + 2HBr \longrightarrow Xe + 2HF + Br_2$$
$$XeF_2 + 2HI \longrightarrow Xe + 2HF + I_2$$

(ii) It oxides BrO_3^- as well as Ce^{+3} into BrO_4^- and Ce^{+4} respectively.

(iii) Due to oxidising property it is used as a fluorinating agent.

$$C_6H_5I + XeF_2 \longrightarrow C_6H_5IF_2 + Xe$$
$$CH_3I + XeF_2 \longrightarrow CH_3IF_2 + Xe$$

(7) Reaction with HF :

$$3XeF_2 + Pt \xrightarrow{HF} PtF_6 + 3Xe$$
$$24XeF_2 + S_8 \longrightarrow 8SF_6 + 24Xe$$

(8) Reaction with Ammonia :

$$3XeF_2 + 8NH_3 \longrightarrow N_2 + 6NH_4F + 3Xe$$

(9) Reaction with Nitrogen oxide :

$$XeF_2 + 2NO_2 \longrightarrow \underset{\text{(nitronium fluoride)}}{2NO_2F} + Xe$$

(10) Formation of addition compounds :

$$XeF_2 + PF_5 \longrightarrow [XeF]^+ [PF_6]^-$$

$$XeF_2 + IF_5 \longrightarrow [XeF]^+ [IF_6]^-$$

$$XeF_2 + 2sbF_5 \longrightarrow [XeF]^+ [Sb_6]^-$$

Fluro anion and cationic species are formed.

Q.141 Which is incorrect statement for XeF_2 –

(A) It has linear structure

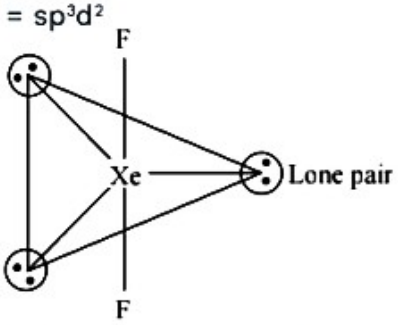
(B) It is hydrolysed rapidly in aqueous solution of a base

(C) It oxidizes Cl^- and I^- to Cl_2 and I_2 respectively

(D) It cannot act as F^- donor

Sol. **(D)** XeF_2

(A) Hybridisation = $\frac{8+2}{2} = 5 = sp^3d^2$



Central atom have three lone pair at equitorial position and two fluorine atoms lie in a straight line thus its shape is linear.

(B) $2XeF_2 + \underset{\text{Base}}{OH^-} \rightarrow 2Xe + 4F^- + 2H_2O + O_2$

(C) $XeF_2 + 2HCl \rightarrow Xe + H_2F_2 + Cl_2$

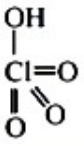
$XeF_2 + 2HI \rightarrow Xe + H_2F_2 + I_2$

(D) Because F is strong oxidising agent thus can act as F^- donar.

$XeF_6 + HF \rightarrow [XeF_5]^+ . [HF_2]^-$

Q.142 **Reference to observation (v) the products of the reaction are ?**

(A) $Xe + F_2 + HClO_4$ **(B)** $XeO_2F_2 + ClO_2$

(C) $Xe(OCl_4)_2 + 2HF$ **(D)** $F - Xe - F : \rightarrow$ 

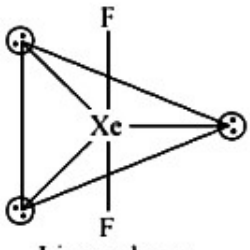
Sol. **(C)** $XeF_2 + 2HClO_4 \rightarrow Xe(OCl_4)_2 + 2HF$

Q.143 **XeF_2 is isostructural with–**

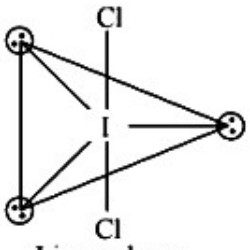
(A) TeF_2 **(B)** ICl_2^- **(C)** $SbCl_3$ **(D)** $BaCl_2$

Sol. **(B)**

XeF_2	ICl_2^-
$= \frac{8+2}{2} = 5$	$= \frac{7+2+1}{2} = 5$
$= sp^3d$	$= sp^3d$



Linear shape



Linear shape

TeF_2	$BaCl_2$	$SbCl_3$
$= \frac{6+2}{2} = sp^3 = (4)$	$\frac{2+2}{2} = sp$	$\frac{5+3}{2} = 4\ sp^3$

2. **Xenon Tetrafluoride (XeF_4) : -**

Preparation :

By taking ratio 1 : 5 of Xe : F_2 give XeF_4.

$$Xe + 2F_2 \xrightarrow[\text{monel tube}]{873K,7bar} XeF_4$$

1 : 5

Properties :

(i) It is colourless crystalline solid

(ii) It sublimes at 298 k and undergoes sublimation, soluble in CF_3COOH.

(iii) It undergoes hydrolysis violently so no moisture must be present during its preparation.

(iv) Reaction with water : (Hydrolysis)

$$6XeF_4 + 12H_2O \longrightarrow 4Xe + 2XeO_3 + 24HF + 3O_2$$

- In above reaction XeO_3 is formed which is white solid and explosive dry compound and water soluble.
- This XeO_3 reacts with NaOH and form sodium xenate.

$$XeO_3 + NaOH \longrightarrow \underset{\text{Sodium xenate}}{Na^+[HXeO_4]^-}$$

- Xenate disproportionate into perxenate ion in basic medium.

$$2[HXeO_4]^- + 2OH^- \longrightarrow [XeO_6]^{-4} + Xe + O_2 + 2H_2O$$

Xenic acid (H_2XeO_4) is very weak acid.

(v) Oxidising properties of XeF_4

$$XeF_4 + 2H_2 \longrightarrow Xe + 4HF$$

$$XeF_4 + 2Hg \longrightarrow Xe + 2HgF_2$$

(vi) Addition Reaction

XeF_4 form cationic & fluor anion by reacting with fluorides.

$$XeF_4 + SbF_5 \longrightarrow [XeF_3]^+[SbF_6]^-$$

Q.144 Which compound is prepared by the following reaction–

$$Xe + 2F_2 \xrightarrow[673\,k,\,5-6\,atm]{\text{Ni vessel}}$$

(A) XeF_2 **(B) XeF_6** **(C) XeF_4** **(D) $XeOF_2$**

Sol. **(C)** $Xe + 2F_2 \xrightarrow[673K,\,5.6\,atm]{Ni\ vessel} \underset{\text{Xenon tetra fluoride}}{XeF_4}$

3. **Xenon hex a fluoride (XeF_6) :**

Preparation :

(i) By taking ration 1 : 20 of Xe : F_2

$$Xe + 3F_2 \xrightarrow[\text{monel tube}]{573K,60-70bar} XeF_6$$

1 : 20

(ii) By reaction with O_2F_2:

$$XeF_4 + O_2F_2 \longrightarrow XeF_6 + O_2$$

Properties :

(i) It is colourless crystalline solid.

(ii) It provides yellow liquid on melting while other give white liquid on melting.

(iii) soluble in HF

(iv) Hydrolysis

a) Complete hydrolysis

$$XeF_6 + 3H_2O \longrightarrow \underset{\text{(white solid)}}{XeO_3} + 2HF$$

b) Partial hydrolysis

$$XeF_6 + H_2O \longrightarrow \underset{\text{(colourless solid)}}{XeOF_6} + 2HF$$

(v) Reaction with silica

$$2XeF_6 + SiO_2 \longrightarrow 2XeOF_4 + SiF_4$$

(vi) Heating effect

By heating hexafluoride give di and tetra fluoride of Xenon.

$$2XeF_6 \xrightarrow{\Delta} XeF_2 + XeF_4 + 3F_2$$

Note : XeF_4 and XeF_2 does not decompose.

(vii) Addition reaction :

$$XeF_6 + SbF_5 \longrightarrow [XeF_2]^+ [SbF_6]^-$$

$$XeF_6 + BF_3 \longrightarrow [XeF_5]^+ [BF_4]^-$$

They form cationic species and house anion by reaction with fluorides.

(viii) Reaction with hydrogen :

$$2XeF_6 + 3H_2 \longrightarrow 3XeOF_4$$

(ix) Reaction with XeO_3:

$$2XF_6 + XeO_3 \longrightarrow 3XeOF_6$$

Q.145 **XeF_6 on complete hydrolysis yields–**

(A) XeO_2F_2 **(B) XeO_2F_4** **(C) XeO_3** **(D) XeO_2**

Sol. **(C)** XeF_6 on complete hydrolysis (by atmospheric moisture) gives the solid XeO_3.

$XeF_6 + 3H_2O \rightarrow XeO_3 + 3H_2F_2$

Q.146 **The compound that attacks pyrex glass is–**

(A) XeF_2 **(B) XeF_4** **(C) XeF_6** **(D) None**

Sol. **(C)** XeF_6 can be sublimed at room temperature it is more susceptible to hydrolysis XeF_6 melt to give yellow liquid, since it attacks pyrex glass. because silica react with XeF_6.

$2\ XeF_6 + 3SiO_2 \rightarrow 3SiF_4 + 2XeO_3$

Q.147 **Which of the following are formed by Xenon –** **[REE - 2000]**

(A) XeF_3 **(B) XeF_4** **(C) XeF_5** **(D) XeF_6**

Sol. **(B,D)**

$$Xe + 2F_2 \xrightarrow[Ni]{400^\circ C} XeF_4$$

$$Xe + 3F_2 \xrightarrow[50\,atm]{300^\circ C} XeF_6$$

By unpairing of one paired orbital. two singly occupied orbitals come into existence thus either two or four, six singly occupied orbitals can be formed instead of one, three or five singly occupied orbitals thus it can't acts as fluorinating agent and doesn't give addition reaction with fluorine thus doesn't form XeF, XeF_3, XeF_5 etc

B. Compounds of Xenon with oxygen :

1. **Xenon tri oxide (XeO_3)**

Preparation :

(i) By hydrolysis of XeF_4

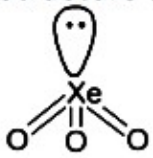
$$6XeF_4 + 12H_2O \longrightarrow 4Xe + 2XeO_3 + 24HF + 3O_2$$

(ii) By hydrolysis of XeF_6

$$XeF_6 + 3H_2O \longrightarrow XeO_3 + 6HF$$

Properties :

(i) XeO_3 is colourless and explosive solid.
(ii) It has pyramidal geometry
(iii) Structure :



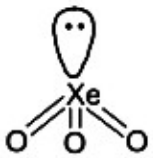
Q.148 XeO_3 is an explosive compound. (T/F)

Sol. True.
It explodes violently, when dry, on gentle heating rubbing or pressing. It produces same effect as T.N.T.

Q.149 In XeO_3, Xe is –

(A) sp^3 - hybridised **(B) sp^2 - hybridised**
(C) sp - hybridised **(D) sp^3d - hybridised**

Sol. (A) Hybridisation



$$= \frac{\text{valence shell } e^- \text{ of central atom} + \text{attached only mono atomic atom}}{2}$$

Diatomic (such as O_2, N_2 not considered)

$$= \frac{\text{Xe valence } e^- = 8 + O_2 \text{not considered} = (0)}{2}$$

$$= \frac{8+0}{2} = 4 = sp^3$$

XeO_3 = sp^3 - hybridised

2. **Oxyfluorides of Xenon :**

Preparation :

By partially hydrolysis process we can get oxyfluorides of Xenon.

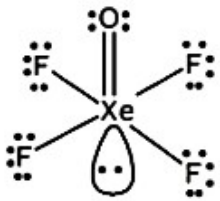
(i) $XeF_6 + H_2O \longrightarrow XeOF_4 + 2HF$

(ii) $XeF_6 + 2H_2O \longrightarrow XeO_2F_2 + 4HF$

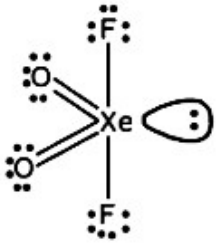
Properties :

(i) $XeOF_4$ is colourless volatile liquid
(ii) Geometry of $XeOF_4$ is square is triagonal pyramidal.
(iii) XeO_2F_2 is triagonal bipyramidal
(iv) XeO_2F_2 is colourless solid.
(v) Structure :

$XeOF_4$



XeO_2F_2



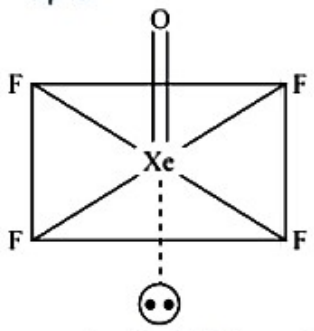
Q.150 **Geometry of $XeOF_4$ molecule is–**

(A) Square planar **(B) Square pyramidal**

(C) Triangular bipyramid **(D) Distorted octahedron**

Sol. **(B)** $XeOF_4$ p bond = 1

Hybridisation = $\frac{8+4+0}{2}$ = 6 = sp^3d^2



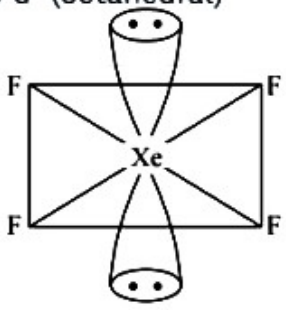
Shape = square pyramidal geometry (sp^3d^2) (one lone pair)

Q.151 **Which of the following is correctly matched–**

(A) XeF_4 .... tetrahedral **(B) XeO_3 ..... trigonal pyramidal**

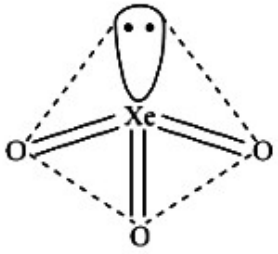
(C) $XeOF_4$ ..... square pyramidal **(D) $XeOF_4$ ..... tetrahedral**

Sol. **(B,C,D)** XeF_4 = $\frac{8+4}{2}$ = 6, sp^3d^2 (octahedral)



shape square planer

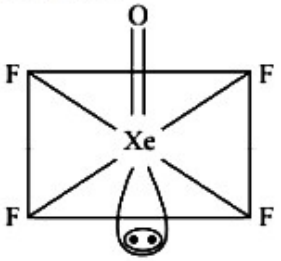
(b) XeO_3 = $\frac{8}{2}$ = sp^3 (due to one lone pair pyramidal shape)



Pyramidal shape
Geometry Tetrahedral

(c) $XeOF_4$ Geometry shape

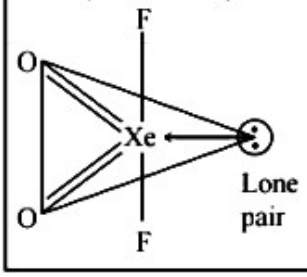
$\frac{8+4}{2} = 6\ sp^3d^2$ octahedral square pyramidal (due to one lone pair)



Lone pair

(d) XeO_2F_2 $\frac{8+2}{2} = 5 = sp^3d$

Hybridization and (structure)	Geometry Trigonal bipyramidal	Shape distorted trigonal bipyramidal



Uses of Noble gases :

1. Helium is light and non-informal
2. Helium gas use for filling balloons
3. Helium is used in gas cooled nuclear reactor
4. Liquid helium used as cryogenic agent for various experiments at low temperature.
5. Helium used for production if powerful super conducting magnets.
6. Helium used as NMR spectrometers and MRI scan.

7. Helium is used in gaseous mixture for scuba divers tank due to low solubility in blood.
8. Neon is used in discharge tubes and sign board (fluorescents bulls for advertisement).
9. Neon bulb are used in botanical gardens & green houses.
10. Argon is used to provide inert atmosphere in high temperature metallurgical processes.
11. Argon is used to filling electric bulbs.
12. Argon used as laboratory purpose.
13. Xenon and krypton are used in light bulbs.

Q.152 **Which of the following is / are correct–**

(A) He is used in gas cooled nuclear reactor.

(B) He is used as a cryogenic agent for carrying out experiments at low temperature.

(C) He is used to produce and sustain super conduction magnets.

(D) He is used to fill gas balloons instead of H_2 because it is lighter and non flamable.

Sol. **(A,B,C)** Because He is non combustible gas